PAULEE GROUP · CENTER FOR QUANTUM TECHNOLOGY, KIST

Quantum Signal Processing for Hyperfine-Conditioned Control

Engineering the operations to the detuning

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2026-10-27 · Lab Meeting

Outline

1. **Double quantum with an unpolarized ^{14}N.**

- The unpolarized nucleus gives three detunings, and one π pulse cannot serve all three.

2. **Quantum signal processing.**

- Pulses and equal waits turn the detuning into a polynomial in terms of detuning

3. **Applications.**
   - Universal NOT on the single-quantum qubit $|0\rangle \leftrightarrow |-1\rangle$.
   - DQ state preparation.
   - DQ π pulse.

INTRO METHOD STRATEGY RESULTS ONGOING

01

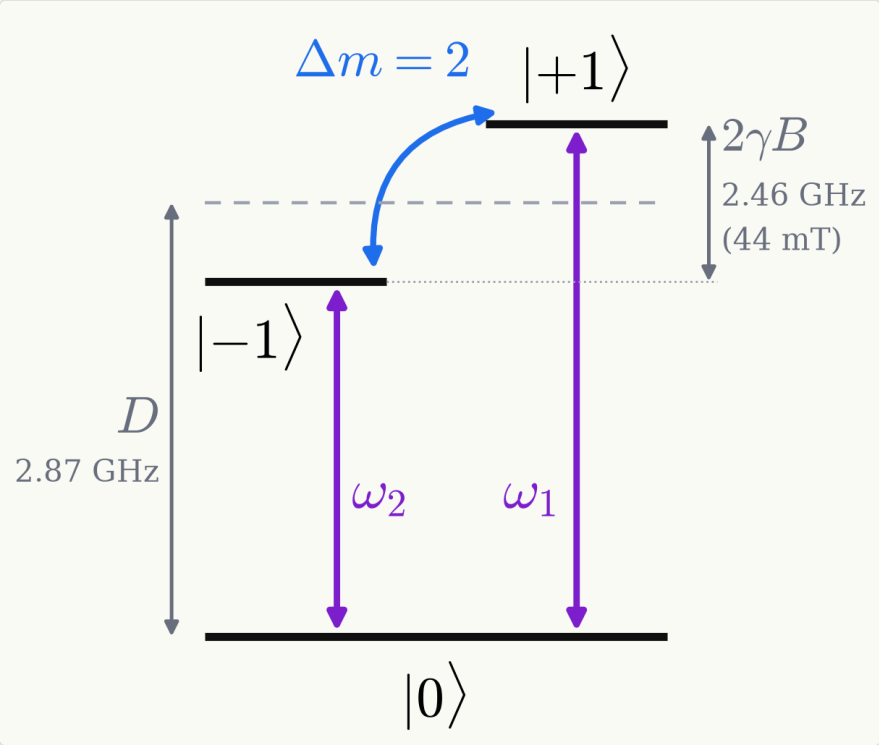
Double quantum with an unpolarised ^{14}N

Two frequency components, three hyperfine manifolds

INTRO

Double-quantum transition

Two frequency components drive the $\Delta m = 2$ transition between $|+1\rangle$ and $|-1\rangle$.



	superposition	phase after t
single quantum	$\|0\rangle + \|+1\rangle$	$\gamma B t$
double quantum	$\|-1\rangle + \|+1\rangle$	$2\gamma B t$

- One frequency component drives each transition $0 \leftrightarrow \pm 1$.
- The ± 1 superposition picks up twice the phase per unit field.
- Twice the phase doubles the magnetometric sensitivity.

Mamin et al., PRL 113, 030803 (2014)

INTRO

Rotating frame

The rotating frame leaves a differential and a common-mode detuning. In the laboratory frame, with a field B along the NV axis and one microwave field with two frequency components along x,

$$H(t) = DS_z^2 + \gamma B S_z + \left[\gamma B_1 \cos(\omega_1 t + \delta_1) + \gamma B_2 \cos(\omega_2 t + \delta_2)\right] S_x .$$

Rotate at ω_1 on $|+1\rangle$ and at ω_2 on $|-1\rangle$, and drop the terms that oscillate at frequencies of order D or $2\gamma B$ (rotating-wave approximation; Mamin et al., PRL 113, 030803 (2014)). In the basis $(|+1\rangle, |0\rangle, |-1\rangle)$,

$$\tilde{H} = \begin{pmatrix} \Delta_1 & \frac{\gamma B_1}{2\sqrt{2}} e^{-i\delta_1} & 0 \\ \frac{\gamma B_1}{2\sqrt{2}} e^{i\delta_1} & 0 & \frac{\gamma B_2}{2\sqrt{2}} e^{i\delta_2} \\ 0 & \frac{\gamma B_2}{2\sqrt{2}} e^{-i\delta_2} & \Delta_2 \end{pmatrix}, \qquad \Delta_{1,2} = D \pm \gamma B - \omega_{1,2}.$$

$$\mathrm{diag}(\Delta_1, 0, \Delta_2) = \Delta\, S_z + \Delta_c\, S_z^2, \qquad \Delta = \tfrac{\Delta_1 - \Delta_2}{2}, \quad \Delta_c = \tfrac{\Delta_1 + \Delta_2}{2}.$$

- The differential detuning Δ comes from the magnetic field and the hyperfine shift.
- The common-mode detuning Δ_c comes from the carrier offset from D, temperature and strain.

INTRO

Spin-1 representation

A balanced drive moves the spin-1 like a spin-½. Equal, mirrored components ($B_1 = B_2$, $\delta_1 = -\delta_2 = \varphi$, $\Delta_c = 0$, $\Omega_{DQ} = \gamma B_{\mathrm{mw}}/2$) give a spin-1 and a spin-½ in the same effective field:

$$\tilde{H} = \Delta S_z + \Omega_{DQ}(\cos\varphi\, S_x + \sin\varphi\, S_y) \quad \longleftrightarrow \quad h = \tfrac{\Delta}{2}\sigma_z + \tfrac{\Omega_{DQ}}{2}(\cos\varphi\,\sigma_x + \sin\varphi\,\sigma_y).$$

The three levels are the symmetric triplet of two spin-½, and $\mathbf{S} = \mathbf{s}_1 + \mathbf{s}_2$ puts the same field on both:

$$|{+1}\rangle = |{\uparrow\uparrow}\rangle, \quad |0\rangle = \tfrac{1}{\sqrt{2}}\big(|{\uparrow\downarrow}\rangle + |{\downarrow\uparrow}\rangle\big), \quad |{-1}\rangle = |{\downarrow\downarrow}\rangle, \qquad \tilde{H} = h \otimes \mathbb{1} + \mathbb{1} \otimes h \;\Rightarrow\; e^{-i\tilde{H}t} = e^{-iht} \otimes e^{-iht}.$$

Equivalently, S_i and $\sigma_i/2$ obey the same commutation relations, so one generator gives one rotation, as a 3×3 or a 2×2 matrix; classically, $\langle \mathbf{S} \rangle$ precesses like a Bloch vector. Hence

$$e^{-i\tilde{H}t} = D^1\big(e^{-iht}\big) \;\text{ for every } \Delta \quad \Longrightarrow \quad \text{a sequence of pulses and waits acts as } D^1(u).$$

INTRO

Spin-1 representation

Two spin-½ in the same field evolve as $u \otimes u$; in the triplet and singlet basis it splits into 3×3 and 1×1 blocks.

$$u = \begin{pmatrix} P & -Q^* \\ Q & P^* \end{pmatrix} \Longrightarrow u \otimes u = \begin{pmatrix} P^2 & -PQ^* & -PQ^* & Q^{*2} \\ PQ & |P|^2 & -|Q|^2 & -P^*Q^* \\ PQ & -|Q|^2 & |P|^2 & -P^*Q^* \\ Q^2 & P^*Q & P^*Q & P^{*2} \end{pmatrix} \quad \text{in } (|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle)$$

$$V = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ 0 & 0 & 1 & 0 \end{pmatrix} \text{ columns } |+1\rangle, |0\rangle, |-1\rangle, |S\rangle \qquad \Longrightarrow \qquad V^\dagger (u \otimes u) V = \left(\begin{array}{ccc|c} P^2 & -\sqrt{2}\,PQ^* & Q^{*2} & 0 \\ \sqrt{2}\,PQ & |P|^2 - |Q|^2 & -\sqrt{2}\,P^*Q^* & 0 \\ Q^2 & \sqrt{2}\,P^*Q & P^{*2} & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right)$$

- The singlet block is $|P|^2 + |Q|^2 = \det u = 1$: the triplet never mixes with it.
- Cutting the 3×3 block gives $D^1(u)$, the NV evolution for every Δ.

INTRO

DQ π pulse and state preparation

Both DQ operations are conditions on the spin-½ amplitudes P (stay) and Q (flip).

Spin-½, $|P|^2 + |Q|^2 = 1$:

$$u = \begin{pmatrix} P & -Q^* \\ Q & P^* \end{pmatrix}.$$

Spin-1: the triplet block of $u \otimes u$,

$$D^1(u) = \begin{pmatrix} P^2 & -\sqrt{2}PQ^* & Q^{*2} \\ \sqrt{2}PQ & |P|^2 - |Q|^2 & -\sqrt{2}P^*Q^* \\ Q^2 & \sqrt{2}P^*Q & P^{*2} \end{pmatrix}.$$

From $|0\rangle$, with $q = |Q|^2$:
$P_0 = (1-2q)^2$, $P_{\pm 1} = 2q(1-q)$.

DQ π pulse, $|\pm 1\rangle \to |\mp 1\rangle$: $P = 0$, e.g. $R_x(\pi)$ with $Q = -i$,

$$D^1(u) = \begin{pmatrix} 0 & 0 & Q^{*2} \\ 0 & -1 & 0 \\ Q^2 & 0 & 0 \end{pmatrix}, \qquad \text{only } Q^2 \text{ matters.}$$

DQ preparation, $|0\rangle \to \frac{1}{\sqrt{2}}\big(|+1\rangle + |-1\rangle\big)$:

$$D^1(u)|0\rangle = \begin{pmatrix} -\sqrt{2}\,PQ^* \\ |P|^2 - |Q|^2 \\ \sqrt{2}\,P^*Q \end{pmatrix}, \qquad f = 4\big[\mathrm{Im}(P^*Q)\big]^2.$$

- $|P|^2 = |Q|^2 = \frac{1}{2}$ empties $|0\rangle$; $\mathrm{Re}(P^*Q) = 0$ fixes the relative phase.
- E.g. $R_x(\frac{\pi}{2})$ prepares $-\frac{i}{\sqrt{2}}\big(|+1\rangle + |-1\rangle\big)$.

INTRO

Hyperfine structure

The ^{14}N couples to the electron mainly through $A_{\parallel} S_z I_z$, which is block diagonal in m_I.

$$H = DS_z^2 + \gamma_e B S_z + A_{\parallel} S_z I_z + A_{\perp}(S_x I_x + S_y I_y) + P I_z^2 - \gamma_n B I_z + f(t)\left(S_x - \frac{\gamma_n}{\gamma_e} I_x\right)$$

term	magnitude ($/2\pi$), at 44 mT	effect on the electron transitions
$A_{\parallel} S_z I_z$	-2.16 MHz	detuning $A_{\parallel} m_I$, comparable to the Rabi frequency
$A_{\perp}(S_x I_x + S_y I_y)$	-2.70 MHz	second-order shifts of 2 to 5 kHz
$P I_z^2$, $-\gamma_n B I_z$	-4.95 MHz, -135 kHz	same for every m_s: negligible
nuclear drive	7 kHz	far off resonance

Secular approximation: drop $A_{\perp}$ and the nuclear drive. Then H commutes with I_z:

$$H = \sum_{m=-1}^{1} H_m \otimes |m\rangle\langle m|, \qquad H_m = DS_z^2 + (\gamma_e B + A_{\parallel} m)\, S_z + f(t)\, S_x + \text{const.}$$

Doherty et al., Phys. Rep. 528, 1 (2013)

INTRO

Hyperfine manifolds

Each hyperfine manifold is a spin-1 with its own detuning. In the rotating frame, with the balanced drive, in the basis $(|+1\rangle, |0\rangle, |-1\rangle)$:

$$\tilde{H}_m = \begin{pmatrix} \Delta_m & \frac{\Omega_{DQ}}{\sqrt{2}} e^{-i\varphi} & 0 \\ \frac{\Omega_{DQ}}{\sqrt{2}} e^{i\varphi} & 0 & \frac{\Omega_{DQ}}{\sqrt{2}} e^{-i\varphi} \\ 0 & \frac{\Omega_{DQ}}{\sqrt{2}} e^{i\varphi} & -\Delta_m \end{pmatrix} = \Delta_m S_z + \Omega_{DQ}(\cos\varphi\, S_x + \sin\varphi\, S_y), \qquad \Delta_m = \Delta_0 + mA.$$

$$h_m = \tfrac{\Delta_m}{2}\sigma_z + \tfrac{\Omega_{DQ}}{2}(\cos\varphi\,\sigma_x + \sin\varphi\,\sigma_y), \qquad U = \sum_m D^1(u_m) \otimes |m\rangle\langle m|.$$

Unpolarised nucleus, $p_m = \frac{1}{3}$, $A/2\pi = 2.16$ MHz:

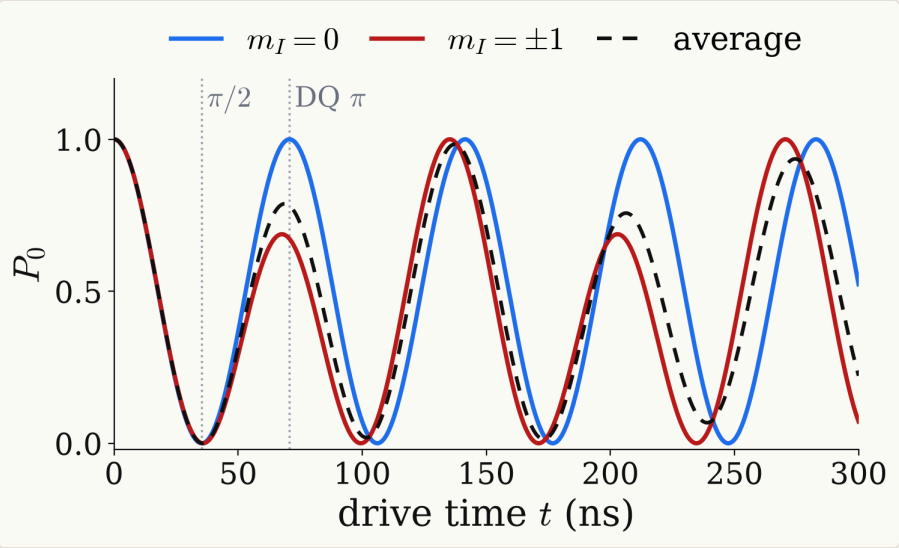
$$\rho_e \longmapsto \frac{1}{3} \sum_{m=-1}^{1} D^1(u_m)\, \rho_e\, D^1(u_m)^\dagger.$$

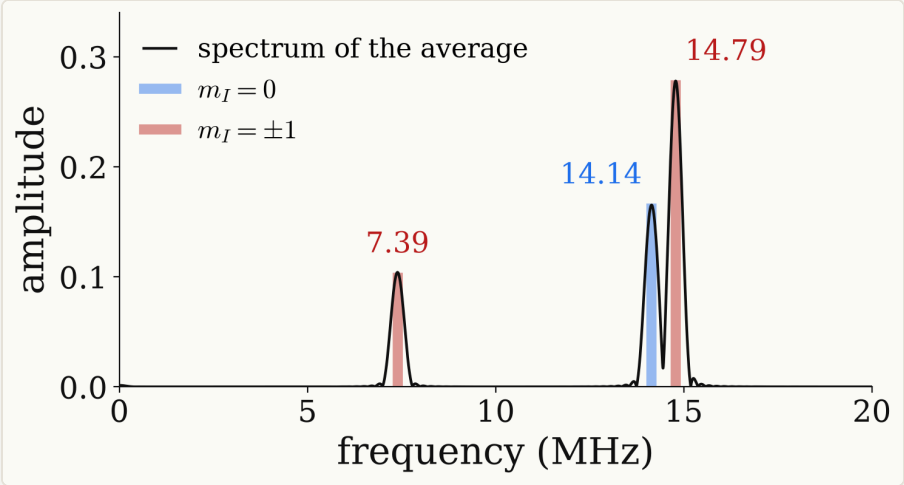
- The drive sits on the $m_I = 0$ resonance, $\Delta_0 = 0$; the outer manifolds see $\pm A$.
- No entanglement is created: the nucleus is a classical label.

INTRO

DQ nutation

The $|0\rangle$ population oscillates differently in each manifold: one line on resonance, two off resonance.





$$P_0^{(m)}(t) = \left[\frac{\Delta_m^2}{\Omega_m'^2} + \frac{\Omega_{DQ}^2}{\Omega_m'^2}\cos\Omega_m' t\right]^2, \quad \Omega_m' = \sqrt{\Omega_{DQ}^2 + \Delta_m^2}$$

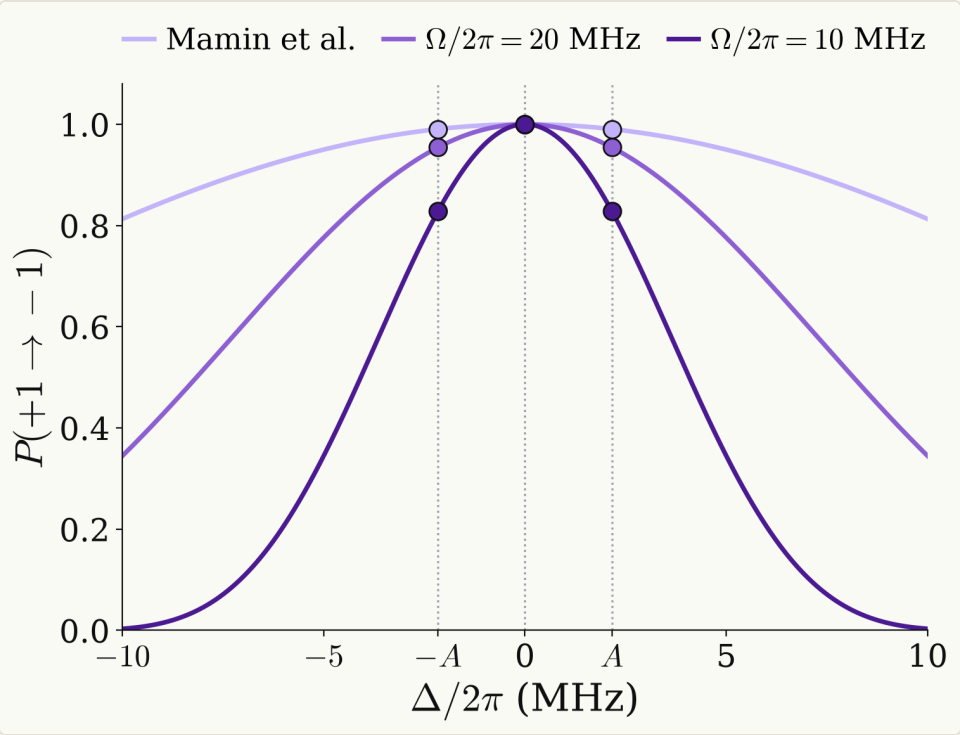
$$P_0^{(m)} = c_m + \frac{2\Delta_m^2\Omega_{DQ}^2}{\Omega_m'^4}\cos\Omega_m' t + \frac{\Omega_{DQ}^4}{2\Omega_m'^4}\cos 2\Omega_m' t$$

- On resonance $P_0 = \cos^2\Omega_{DQ} t$: a single line at $2\Omega_{DQ}$.
- Off resonance $q < 1$: P_0 returns only to 0.69 at the DQ π time, and a line at Ω_m' appears.

INTRO

Single-pulse error

One rectangular DQ π pulse misses the outer hyperfine manifolds.



$$q = 1 - \eta_m^2 + O(\eta^4), \qquad 1 - P_{-1} \approx \tfrac{4}{3}\,\eta_A^2$$

with $\eta_m = \Delta_m/\Omega_{DQ}$ and $\Omega = \sqrt{2}\,\Omega_{DQ}$.

drive	η_A	$1 - P_{-1}$	$\pi/2$ phase
Mamin et al.	0.069	6.3×10^{-3}	$\pm 7.9^\circ$
$\Omega/2\pi = 20$ MHz	0.153	3.0×10^{-2}	$\pm 17.5^\circ$
$\Omega/2\pi = 10$ MHz	0.305	0.115	$\pm 34.9^\circ$

- The π pulse leaves population in $|0\rangle$.
- The $\pi/2$ pulse fails through its phase.

Phase coherence needs one common Q^2: $Q(\Delta_m) = s_m\, e^{i\chi}$ with signs $s_m = \pm 1$ and one phase χ.

02

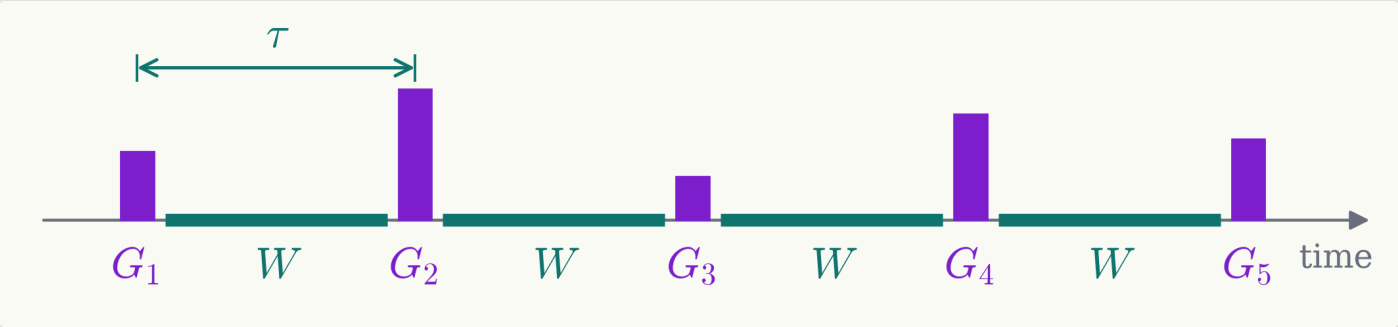
Quantum signal processing

Pulses and waits make polynomials in terms of detuning

METHOD

QSP sequence

A product of pulses and waits is one SU(2) matrix.



$$u = R_x(\phi_{K+1})\, R_z(\Delta\tau)\, R_x(\phi_K) \cdots R_z(\Delta\tau)\, R_x(\phi_1) \;\in\; \mathrm{SU}(2)$$

Any SU(2) matrix is a rotation, $u = e^{-i\alpha\, \mathbf{n}\cdot\boldsymbol{\sigma}} = \cos\alpha\, \mathbb{1} \; - i \sin\alpha\, (n_x X + n_y Y + n_z Z)$:

$$u = \begin{pmatrix} \cos\alpha - i n_z \sin\alpha & -(n_y + i n_x)\sin\alpha \\ (n_y - i n_x)\sin\alpha & \cos\alpha + i n_z \sin\alpha \end{pmatrix} = \begin{pmatrix} P & -Q^* \\ Q & P^* \end{pmatrix}, \qquad |P|^2 + |Q|^2 = 1.$$

- Only the waits depend on the detuning, so P and Q are functions of $\theta = \Delta\tau/2$.

METHOD

Free evolution

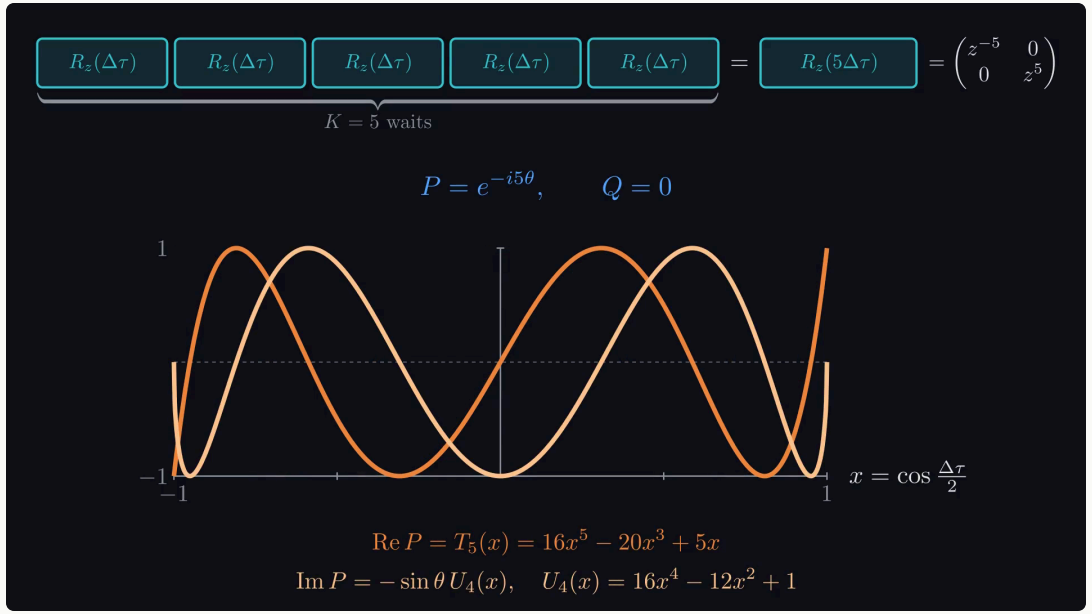
Start with all pulses off, $\phi_k = 0$: K waits make one rotation.

$$u = R_z(K\Delta\tau) = \begin{pmatrix} z^{-K} & 0 \\ 0 & z^K \end{pmatrix}, \qquad z = e^{i\theta}, \quad \theta = \frac{\Delta\tau}{2}$$

$$\mathrm{Re}\, P = \cos K\theta, \qquad \mathrm{Im}\, P = -\sin K\theta = -\sin\theta \frac{\sin K\theta}{\sin\theta}, \qquad Q = 0$$

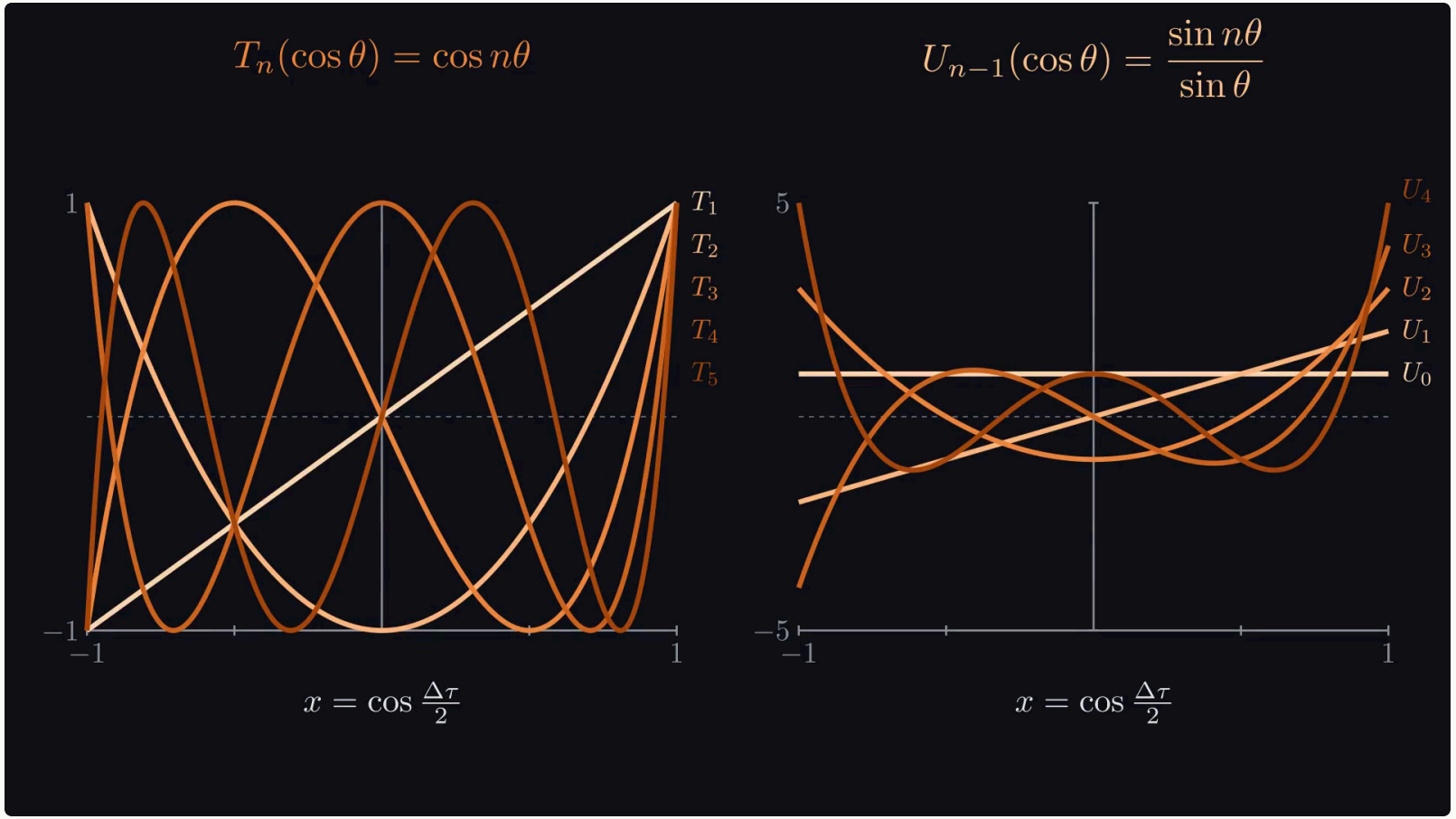
K	$\mathrm{Re}\, P$	$\mathrm{Im}\, P$
1	$\cos\theta$	$-\sin\theta \cdot 1$
2	$2\cos^2\theta - 1$	$-\sin\theta \cdot 2\cos\theta$
3	$4\cos^3\theta - 3\cos\theta$	$-\sin\theta\,(4\cos^2\theta - 1)$

- $\mathrm{Re}\, P$ and $\mathrm{Im}\, P / \sin\theta$ are polynomials in $\cos\theta$, of degree K and $K-1$.



METHOD

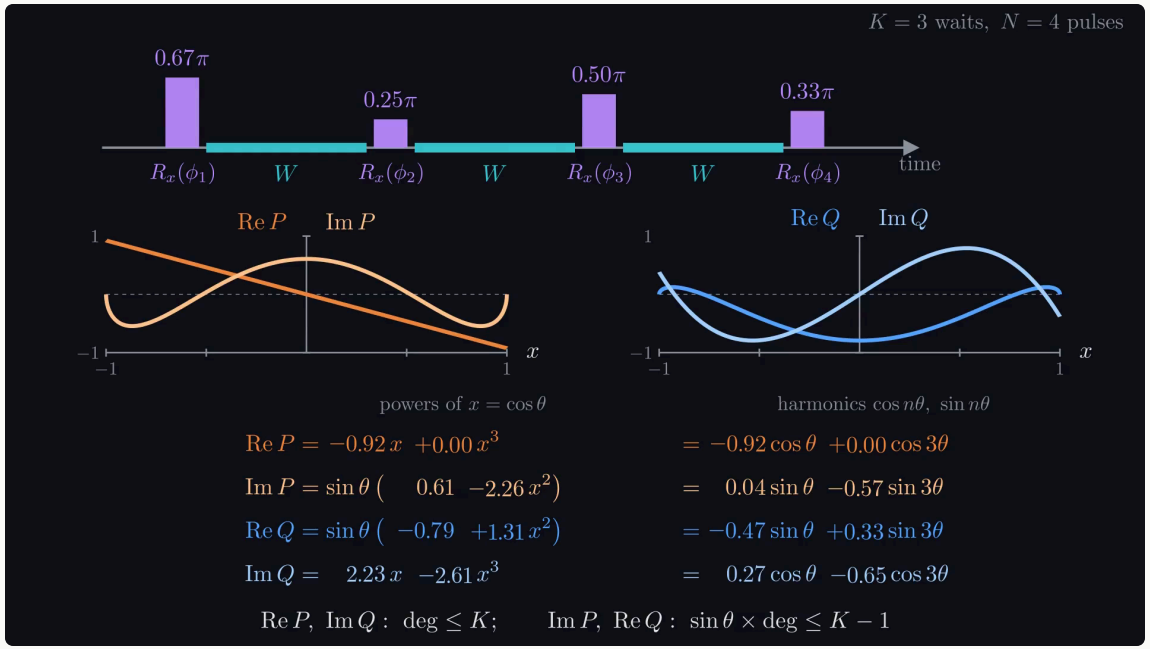
Chebyshev polynomials



METHOD

Pulse angles

Varying $\{\phi_k\}_{K+1}$ tunes $\mathrm{Re}\,P,\ \mathrm{Im}\,P,\ \mathrm{Re}\,Q,\ \mathrm{Im}\,Q$; the degree stays K.



> **The starting idea of QSP:** find $\{\phi_k\}_{K+1}$ that make a target function (which ones: Part III).

INTRO

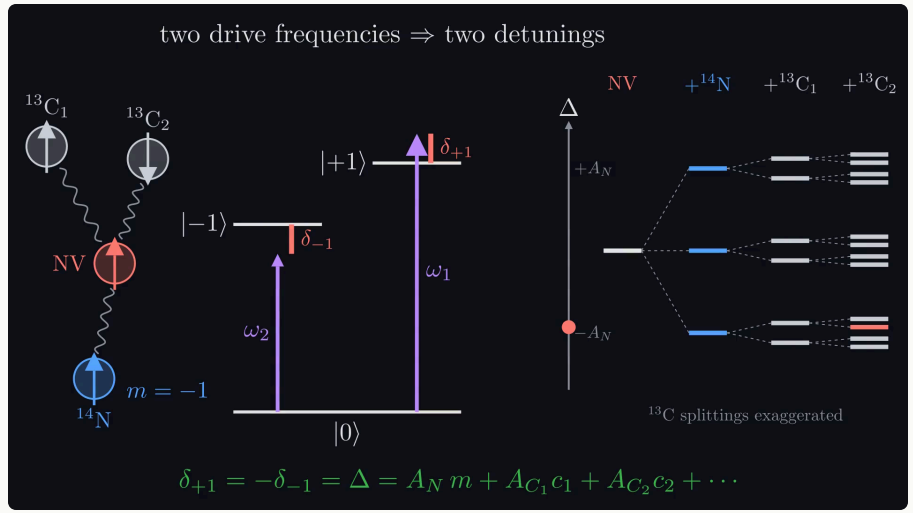
Detuning problem

From the ideal Hamiltonian, with one microwave field of two frequency components,

$$H(t) = DS_z^2 + \gamma B S_z + \left[\gamma B_1 \cos(\omega_1 t + \varphi_1) + \gamma B_2 \cos(\omega_2 t + \varphi_2)\right] S_x,$$

in the double rotating frame, with the rotating-wave and secular approximations, $B_1 = B_2$, $\varphi_1 = -\varphi_2 = \varphi$ and zero detuning, in the basis $(|+1\rangle, |0\rangle, |-1\rangle)$:

$$\tilde{H}_0 = \begin{pmatrix} 0 & \frac{\gamma B_1}{2\sqrt{2}} e^{-i\varphi} & 0 \\ \frac{\gamma B_1}{2\sqrt{2}} e^{i\varphi} & 0 & \frac{\gamma B_1}{2\sqrt{2}} e^{-i\varphi} \\ 0 & \frac{\gamma B_1}{2\sqrt{2}} e^{i\varphi} & 0 \end{pmatrix} = \Omega_{DQ} \left(\cos\varphi\, S_x + \sin\varphi\, S_y\right)$$



- In the presence of interacting spins, including ^{14}N and ^{13}C, each spin induces a detuning that depends on its own spin state.
- Two drive frequencies give two detunings, δ_{+1} and δ_{-1}. A coupling $\propto S_z$ shifts $|\pm 1\rangle$ oppositely: $\delta_{+1} = -\delta_{-1} = \Delta$.

INTRO

Detuning sources

^{14}N. The secular hyperfine term is block diagonal in m_I, and in each block it acts as a detuning:

$$\tilde{H} = \sum_m \tilde{H}_m \otimes |m\rangle\langle m|, \qquad \tilde{H}_m = \tilde{H}_0 + A_N\, m\, S_z = \begin{pmatrix} A_N m & \frac{\gamma B_1}{2\sqrt{2}} e^{-i\varphi} & 0 \\ \frac{\gamma B_1}{2\sqrt{2}} e^{i\varphi} & 0 & \frac{\gamma B_1}{2\sqrt{2}} e^{-i\varphi} \\ 0 & \frac{\gamma B_1}{2\sqrt{2}} e^{i\varphi} & -A_N m \end{pmatrix}$$

Many sources. Every secularly coupled spin adds one label and one term to the detuning:

$$\tilde{H} = \sum_{m, c_1, c_2, \ldots} \tilde{H}_{m c_1 c_2 \ldots} \otimes |m\rangle\langle m| \otimes |c_1\rangle\langle c_1| \otimes |c_2\rangle\langle c_2| \otimes \cdots, \qquad \tilde{H}_{m c_1 c_2 \ldots} = \begin{pmatrix} \Delta & \frac{\gamma B_1}{2\sqrt{2}} e^{-i\varphi} & 0 \\ \frac{\gamma B_1}{2\sqrt{2}} e^{i\varphi} & 0 & \frac{\gamma B_1}{2\sqrt{2}} e^{-i\varphi} \\ 0 & \frac{\gamma B_1}{2\sqrt{2}} e^{i\varphi} & -\Delta \end{pmatrix}$$

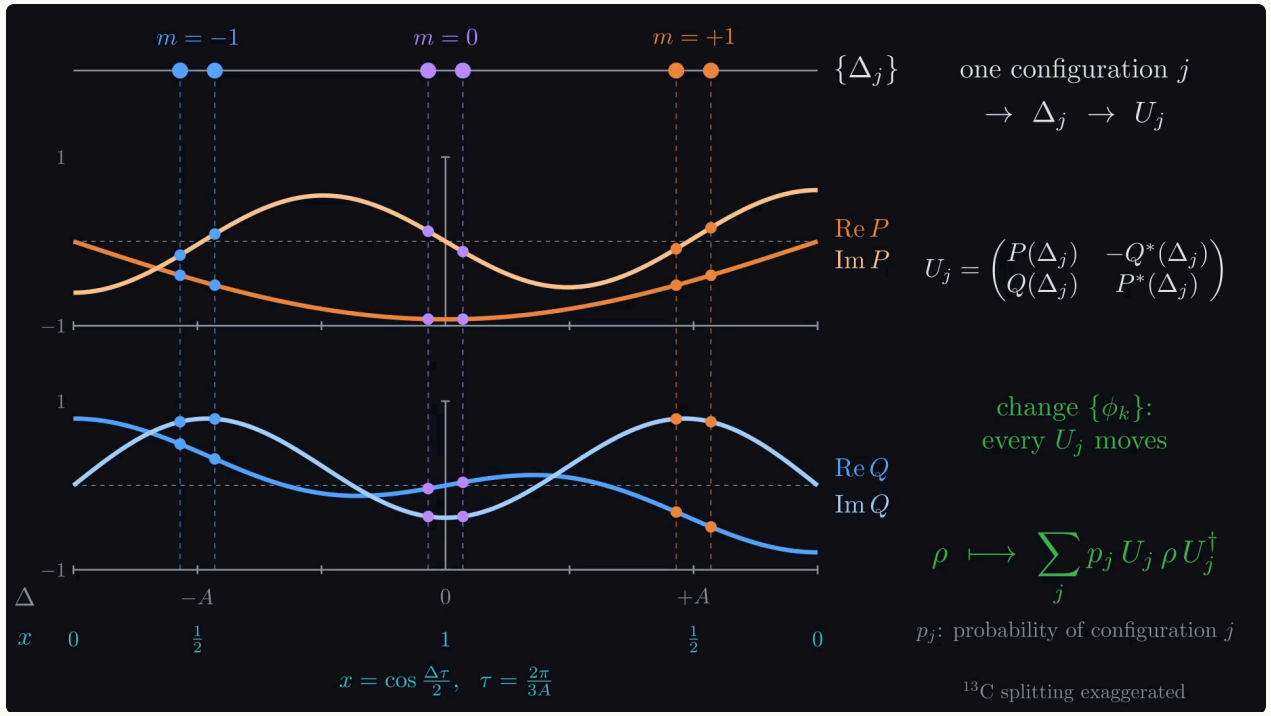
$$\Delta = A_N\, m + A_{C_1} c_1 + A_{C_2} c_2 + \cdots, \qquad A_N/2\pi = 2.16\ \text{MHz}\ (^{14}\text{N}), \quad A_{C_j}/2\pi \sim 100\ \text{kHz}\ (^{13}\text{C})$$

Doherty et al., Phys. Rep. 528, 1 (2013)

STRATEGY

Engineering the response

The idea of QSP is engineering the response $U_{mc_1c_2\ldots} = e^{-i\tilde{H}_{mc_1c_2\ldots}t}$ at every detuning the environment produces.



$$\{\Delta\} \longrightarrow \{\Delta\tau\} \longrightarrow \{x = \cos\tfrac{\Delta\tau}{2}\}$$

The electron sees a mixture over the configurations:

$$\rho \longmapsto \sum_{m,c_1,c_2,\ldots} p_{mc_1c_2\ldots}\, U_{mc_1c_2\ldots}\, \rho\, U^\dagger_{mc_1c_2\ldots}$$

Next steps:

1. **Set the target function**, on purpose: what P and Q must be at each detuning.
2. **Find** $\{\phi_k\}_{K+1}$: the proper K, the proper τ, and how to fit.

03

Applications

Universal NOT on the SQ, DQ state preparation, and the DQ π pulse

STRATEGY

Design steps

The same five steps for each application, first with instantaneous pulses and then with finite pulses.

step	question	tool
1. Target	what must P and Q be on the bands $\Delta_m \pm 3\sigma$?	the physics of the operation; free between the bands
2. Degree K	how many waits does the target need?	Chebyshev decomposition; conditions against parameters
3. Wait τ	where do the lines land on the circle $e^{i\Delta\tau}$?	no overlap: different targets need different signals
4. Solve	which angles $\{\beta_k\}$ and phases $\{\varphi_k\}$?	damped Newton iteration with the exact Jacobian
5. Evaluate	how good is it on the bands?	gate or state fidelity averaged over the ^{13}C shift and the three lines

- Each ^{14}N line $\Delta_m = A_{\parallel} m_I$ is broadened by the ^{13}C bath, rms $\sigma/2\pi = 0.1$ MHz; no detuning source lies between the lines.
- A finite pulse lasts $|\beta|/\Omega$ ($\Omega_{\mathrm{DQ}} = \Omega/\sqrt{2}$ for the DQ qubit) and depends on Δ itself, so the design is solved again with finite pulses.

INTRO

Universal NOT

The universal NOT flips every qubit state to its orthogonal state.

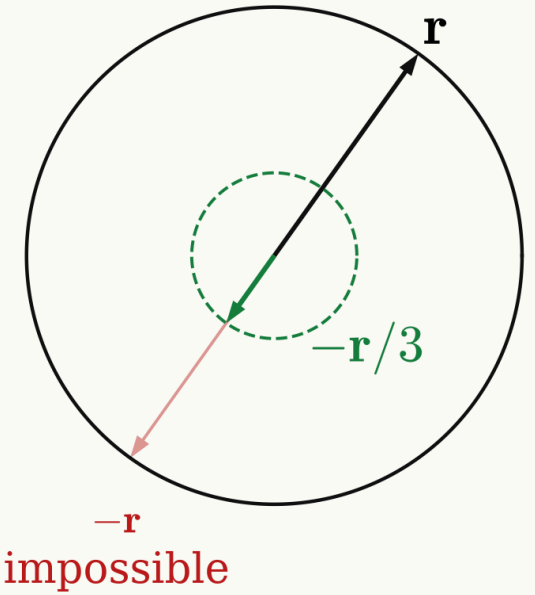
Ideal: $|\psi\rangle \mapsto |\psi^{\perp}\rangle$ for every $|\psi\rangle$, i.e. $\mathbf{r} \mapsto -\mathbf{r}$ on the Bloch sphere.

- Inverting every Bloch vector is a point reflection: antiunitary, so no physical operation does it.

Optimal: the best channel for all inputs at once,

$$\rho \mapsto \frac{1}{3}\sum_{i=x,y,z} \sigma_i \rho \sigma_i = \frac{2\mathbb{1} - \rho}{3}, \qquad \mathbf{r} \mapsto -\frac{\mathbf{r}}{3}, \qquad F = \frac{2}{3}.$$

- Three π rotations about orthogonal axes, applied at random with equal weights.
- The unpolarised ^{14}N supplies that random choice: one rotation per hyperfine line.



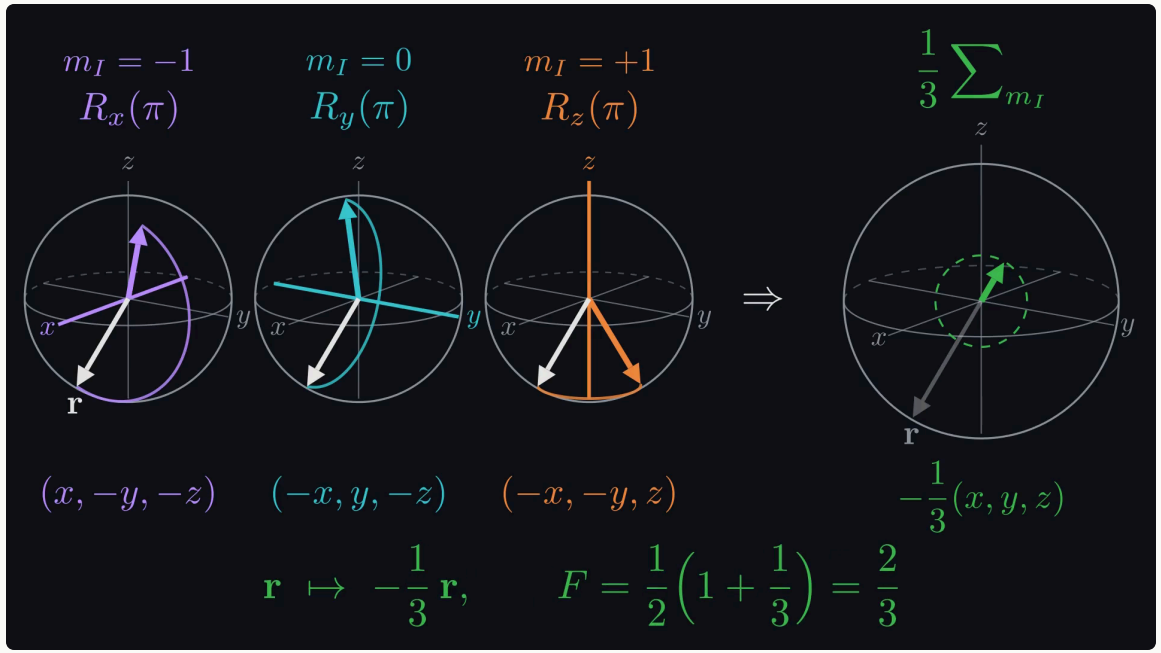
Bužek, Hillery, Werner, PRA 60, R2626 (1999); De Martini, Bužek, Sciarrino, Sias, Nature 419, 815 (2002)

INTRO

Universal NOT: goal

One pulse sequence on the qubit $\{|0\rangle, |-1\rangle\}$ should apply a different π rotation on each ^{14}N line.

$$m_I = -1 :\ R_x(\pi), \qquad m_I = 0 :\ R_y(\pi), \qquad m_I = +1 :\ R_z(\pi) \qquad \Longrightarrow \qquad \frac{1}{3}\sum_{m_I} u_m\, \rho\, u_m^{\dagger} :\ \ \mathbf{r} \mapsto -\tfrac{1}{3}\,\mathbf{r}$$



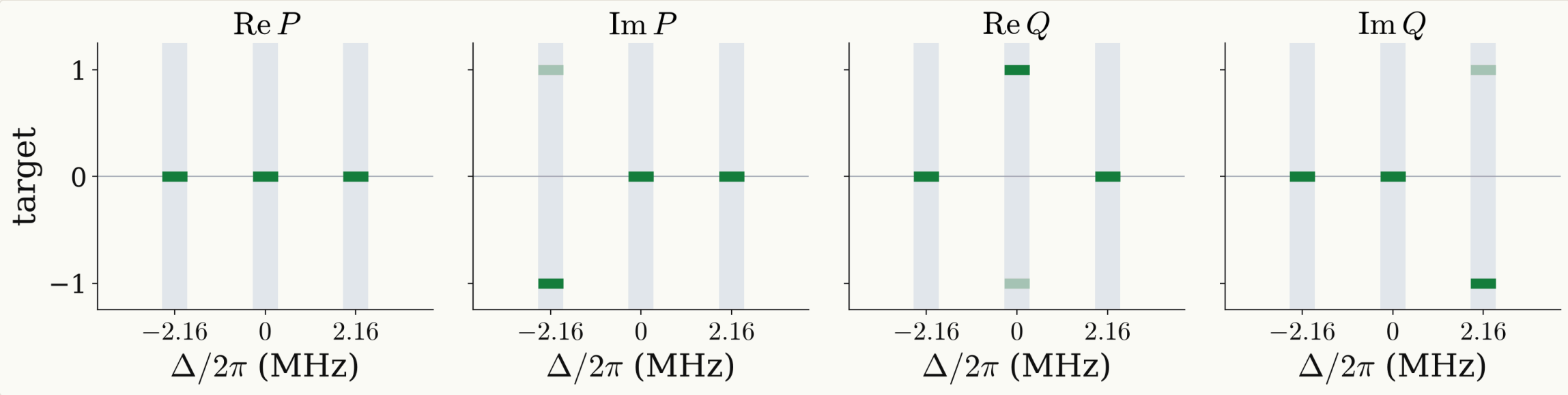
- Every input ends at $F = 2/3$ with its orthogonal state, the optimum.
- One π pulse gives nearly the same rotation on all three lines.

Bužek, Hillery, Werner, PRA 60, R2626 (1999)

STRATEGY

Universal NOT: target function

The target fixes P and Q on the three bands $\Delta_m \pm 3\sigma$ and leaves them free in between.



m_I	$\Delta_m/2\pi$	gate	P	Q
$+1$	-2.16 MHz	$R_z(\pi) = -iZ$	$-i$	0
0	0	$R_y(\pi) = -iY$	0	1
-1	$+2.16$ MHz	$R_x(\pi) = -iX$	0	$-i$

$$F_m = \frac{|\mathrm{Tr}\, T_m^\dagger u|^2 + 2}{6}$$

- Light bars show $-T_m$, which is the same gate.
- Each band has the rms ^{13}C width $\sigma/2\pi = 0.1$ MHz.

METHOD

Universal NOT: pulse phases

Two symmetries rule out the simplest sequences, so the pulses carry phases φ_k and the sequence is arbitrary.

Pulses about $\pm x$: mirror symmetry

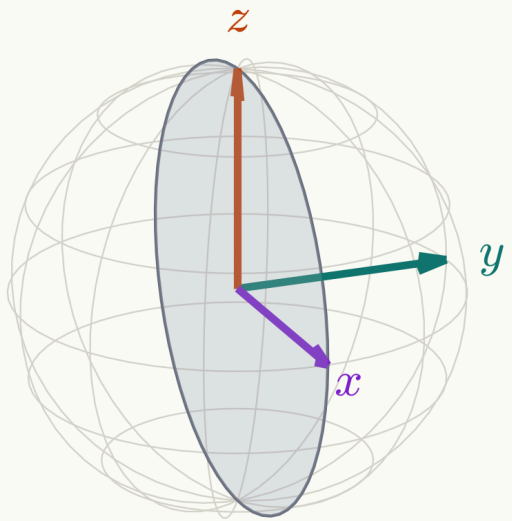
$$X\,h(\Delta)\,X = h(-\Delta) \quad\Longrightarrow\quad u(-\Delta) = X\,u(\Delta)\,X, \qquad u(0) = R_x\Big(\sum_k \beta_k\Big)$$

- $R_z(\pi)$ on $m_I = +1$ forces $XR_z(\pi)X = -R_z(\pi)$, the same gate, on $m_I = -1$, and $m_I = 0$ gets an x rotation.

Symmetric QSP: palindromic (β_k, φ_k), with $\varphi \to -\varphi$

$$R_\varphi(\beta)^T = R_{-\varphi}(\beta) \;\Longrightarrow\; u^T = u \;\Longrightarrow\; Q = -Q^*, \qquad Q = n_y - i\,n_x \;\Longrightarrow\; n_y = 0$$

- All three axes would lie in the xz plane; orthogonal axes cannot.



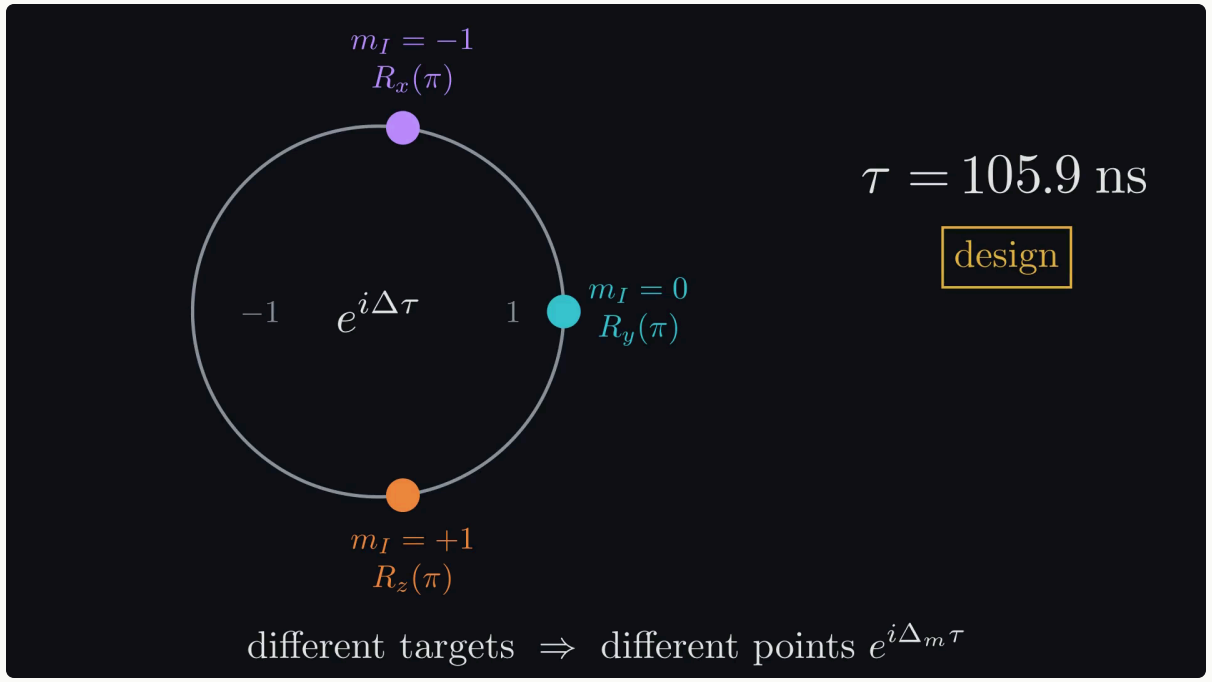
Symmetric QSP: Dong, Meng, Whaley, Lin, PRA 103, 042419 (2021); Wang, Dong, Lin, Quantum 6, 850 (2022)

STRATEGY

Universal NOT: proper τ

Different targets need different signals: no two lines may land on the same point $e^{i\Delta_m \tau}$.

$$u(\Delta) \text{ depends on } \Delta \text{ only through } e^{i\Delta\tau} \text{ (up to sign)} \quad \Longrightarrow \quad e^{iA\tau} \neq 1,\ e^{2iA\tau} \neq 1 \iff A\tau \notin \pi\mathbb{Z}$$



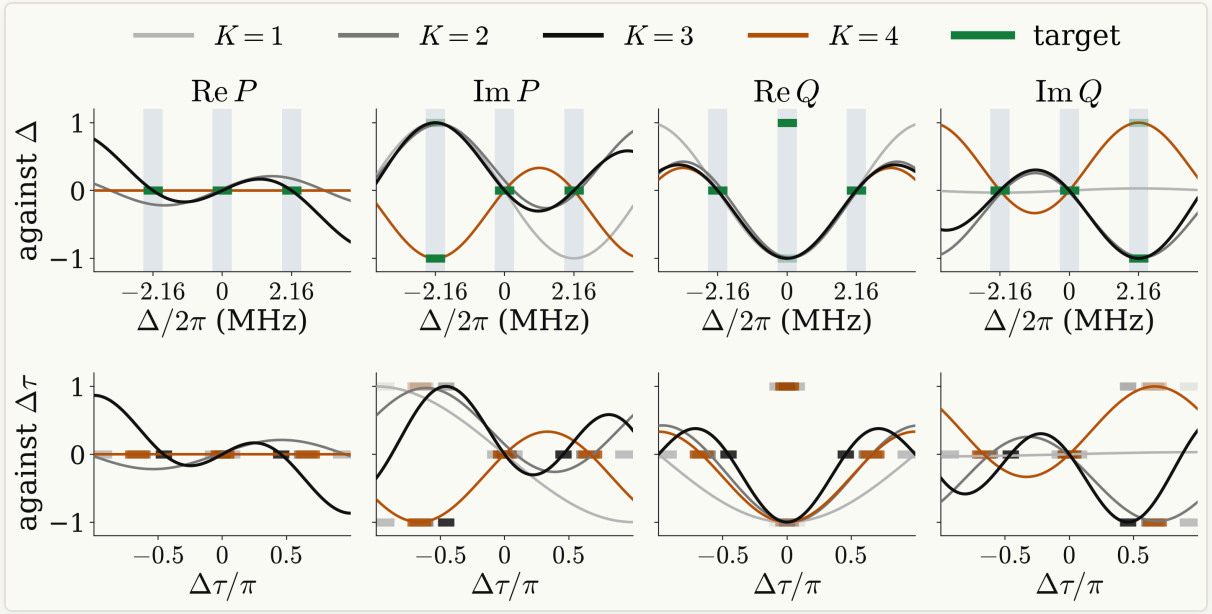
- Not allowed: 231.5 ns ($m_I = \pm 1$ meet) and 463.0 ns (all three meet).
- Pulses about x alone see only $\cos\frac{\Delta\tau}{2}$, equal for $\pm A$: hence the pulse phases.
- Any other τ works; the design uses 105.9 ns.

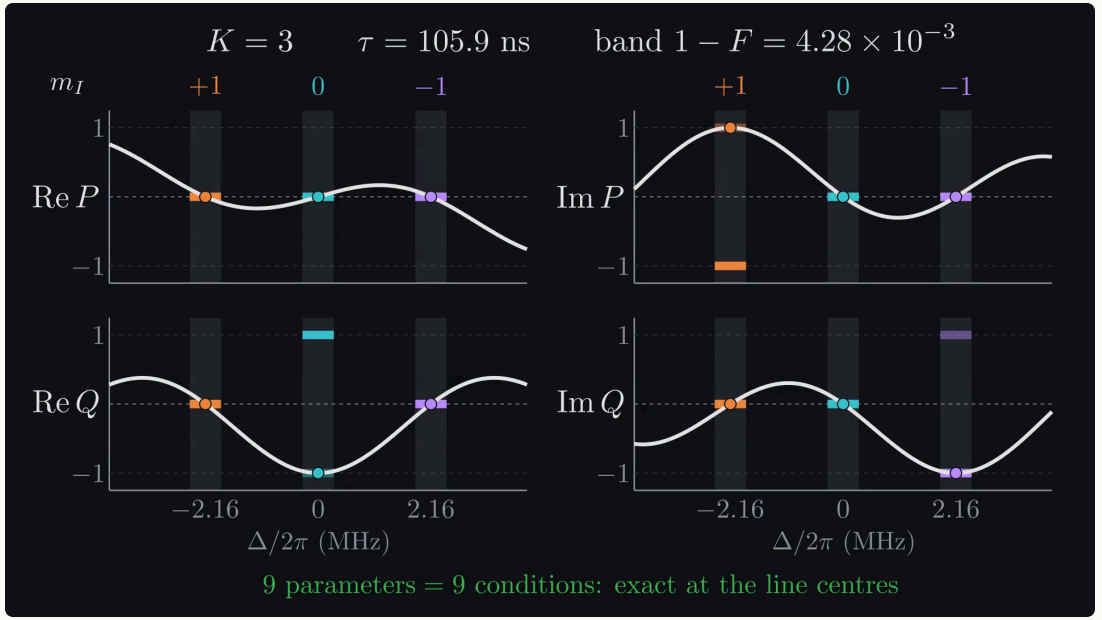
STRATEGY

Universal NOT: proper K

Counting conditions against parameters gives the degree; the response reaches all three targets from $K = 3$ on.

$$3 \times 3 \text{ conditions} \;\le\; 2(K+1)+1 \;\; (\beta_k, \varphi_k, \tau) \;\Longrightarrow\; K = 3; \qquad \text{fixed } \tau_{\text{sep}}: \; 9 \le 2(K+1) \;\Longrightarrow\; K = 4$$





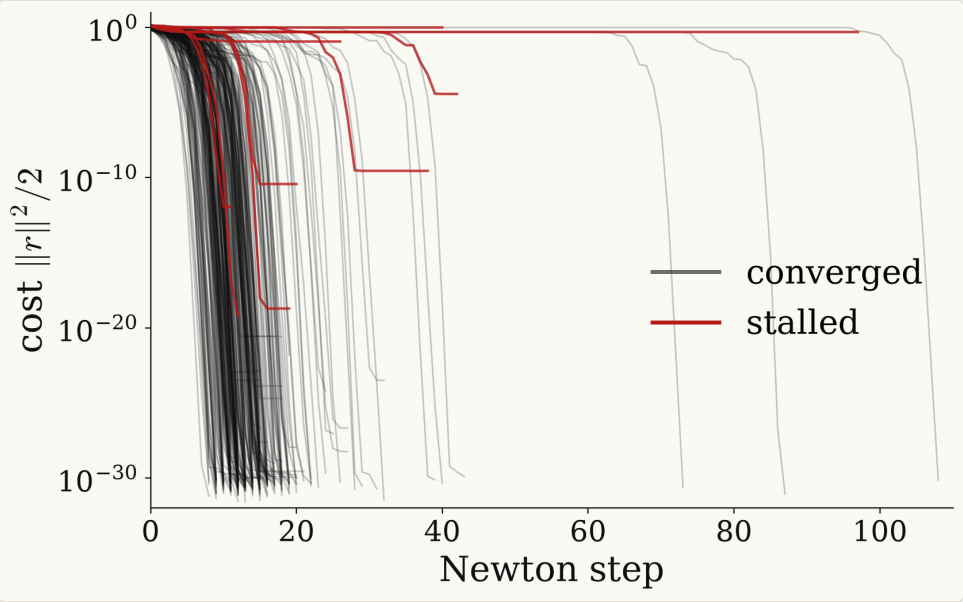
- $K = 1$ and 2 leave $1 - F = 0.22$ and 2.8×10^{-2}.
- $K = 3$ is exact at $\tau = 105.9$ ns.

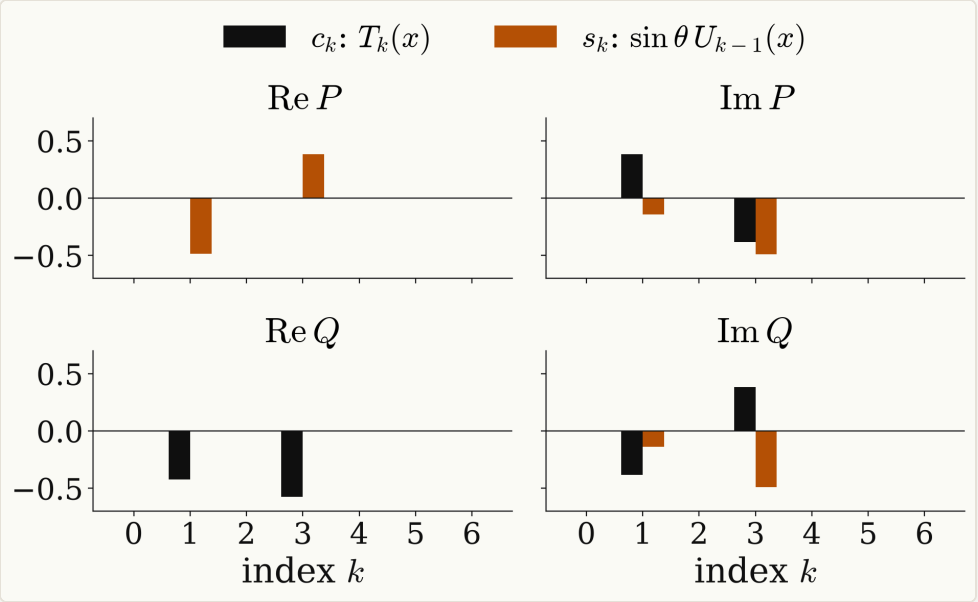
METHOD

Universal NOT: Newton solver

Damped Newton iteration solves the nine equations $r(\mathbf{p}) = 0$ from random starts, with an exact Jacobian.

$$\mathbf{p} \leftarrow \mathbf{p} - \left(J^T J + \lambda \mathbb{1}\right)^{-1} J^T r, \qquad \frac{\partial u}{\partial p_k} = U_N \cdots U_{k+1} \frac{\partial U_k}{\partial p_k} U_{k-1} \cdots U_1$$





- Most starts converge in about 20 steps; a few stall.
- Only odd $k \leq 3$ appear, as degree 3 requires.

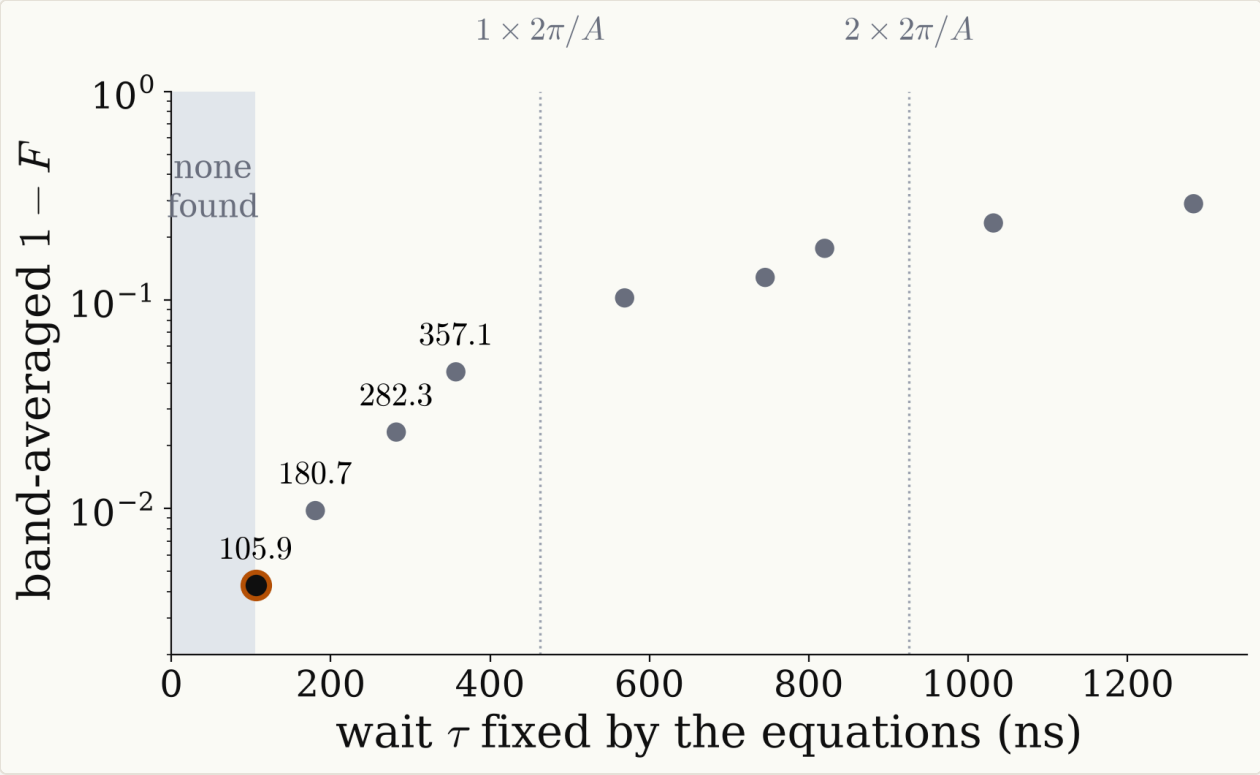
Newton iteration for symmetric QSP: Dong, Lin, Ni, Wang, SIAM J. Sci. Comput. 46, A2951 (2024)

RESULTS

Universal NOT: why $\tau = 105.9$ ns

τ is not tuned: with nine equations for nine unknowns it is an output, and 105.9 ns is the shortest root.

$$r(\beta_1..\beta_4,\ \varphi_1..\varphi_4,\ \tau) = 0 : \quad 9 \text{ equations},\ 9 \text{ unknowns} \implies \text{isolated roots} \quad \tau \in \{105.9,\ 180.7,\ 282.3,\ 357.1\}\ \text{ns} + n\,\tfrac{2\pi}{A}$$



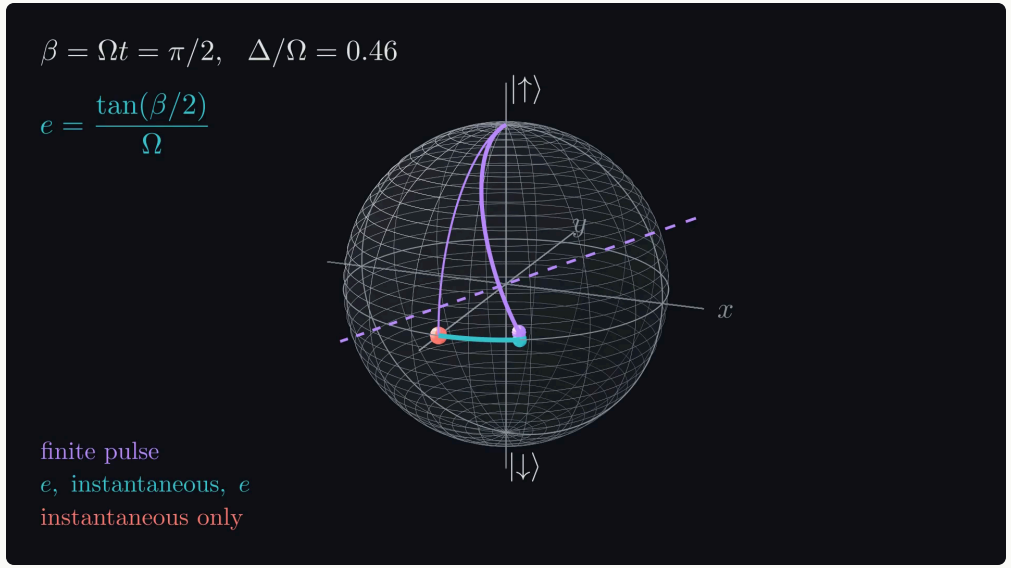
- Every root is exact on the three line centres; only the ^{13}C band error differs.
- A ^{13}C shift δ moves each line by $\delta\tau$ on the circle $e^{i\Delta\tau}$, so the band error grows with τ.
- The shortest root wins: 4.3×10^{-3}, against 9.8×10^{-3} at 180.7 ns.
- Roots pair up as τ and $2\pi/A - \tau$ ($105.9 + 357.1 = 463.0$ ns).

METHOD

Universal NOT: finite pulses

A rectangular pulse acts as a hard pulse between two short free evolutions, so the hard design no longer fits exactly.

$$e^{-i\frac{\beta}{2\Omega}(\Delta Z+\Omega X)} \;\simeq\; R_z(\Delta t_e)\, R_x(\beta)\, R_z(\Delta t_e), \qquad t_e = \frac{\tan(\beta/2)}{\Omega}$$



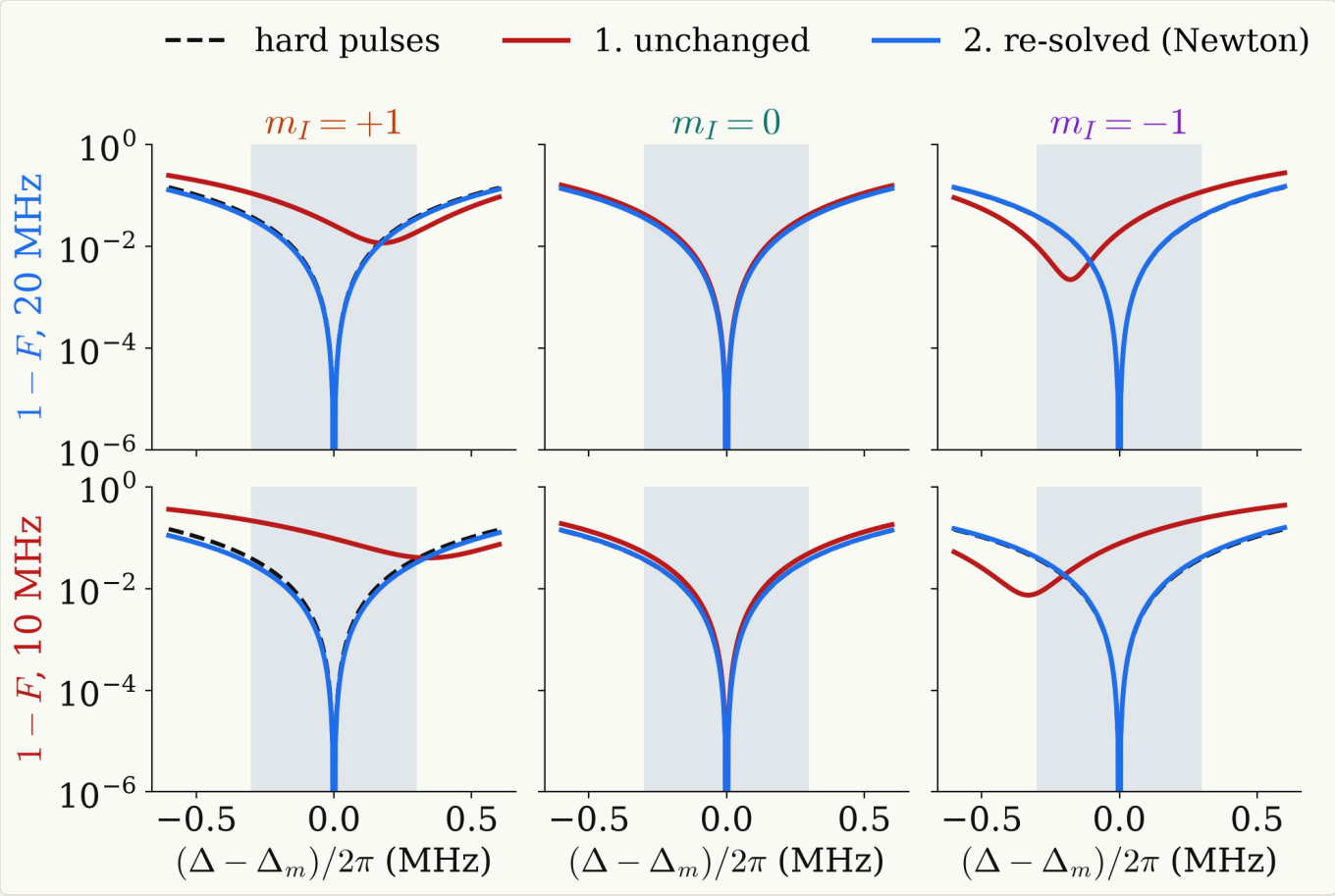
Two ways to play the $K = 3$ design with pulses of length $|\beta_k|/\Omega$:

1. **Unchanged:** the same angles and phases, a wait τ between pulse edges.
2. **Re-solved:** the nine equations solved again by Newton with the finite-pulse propagator, starting from the hard design.

RESULTS

Universal NOT: finite-pulse results

Unchanged, the dip leaves the centre of the $m_I = \pm 1$ lines; re-solving puts it back.



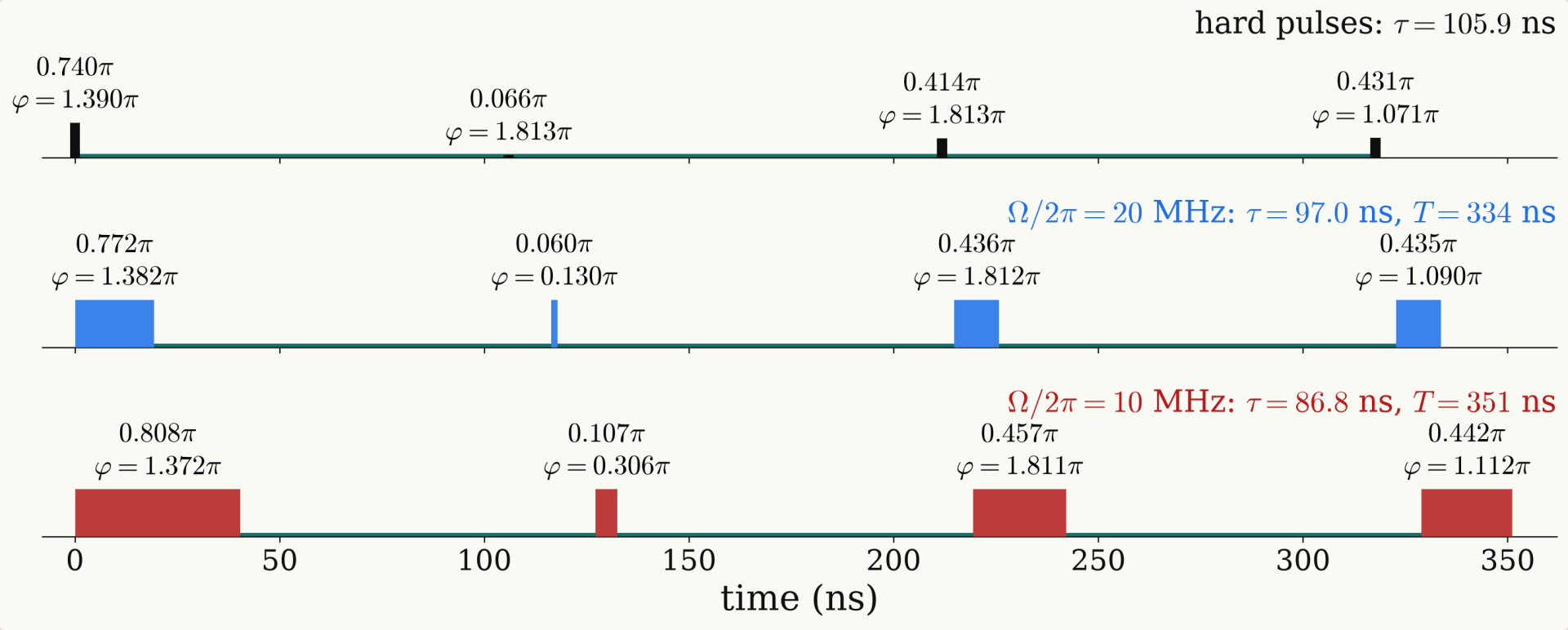
band $1 - F$	20 MHz	10 MHz
1. unchanged	2.0×10^{-2}	6.2×10^{-2}
2. re-solved	4.2×10^{-3}	4.1×10^{-3}

- Re-solved: $\tau = 97.0$ / 86.8 ns, $T = 334$ / 351 ns, exact on all three lines again.

RESULTS

Universal NOT: pulse sequence

The finite-pulse designs stay close to the hard design, with larger main angles and a shorter τ.



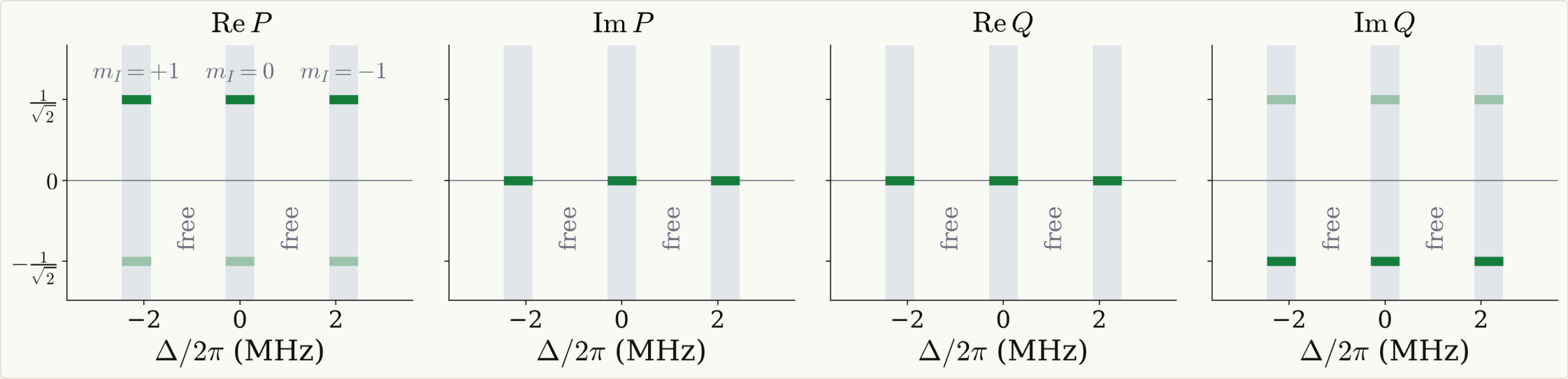
- The refined sequences last $T = 334$ ns at 20 MHz and 351 ns at 10 MHz.
- The second pulse is short: 0.060π lasts 1.5 ns at 20 MHz, comparable to typical rise times.

STRATEGY

DQ preparation: target function

The target is a state, $|0\rangle \to \frac{1}{\sqrt{2}}\big(|{+1}\rangle + |{-1}\rangle\big)$, with the same relative phase in all three manifolds.

$$D^1(u)|0\rangle = \big(-\sqrt{2}\,PQ^*,\ |P|^2 - |Q|^2,\ \sqrt{2}\,P^*Q\big), \qquad f = 4\left[\mathrm{Im}\,(P^*Q)\right]^2$$



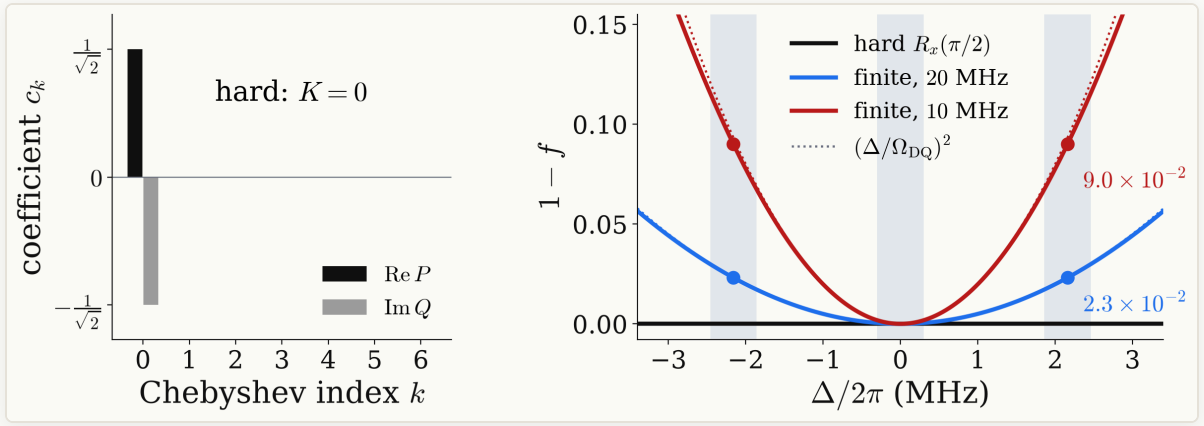
- $f = 1$ if and only if $|P|^2 = \frac{1}{2}$ and $\mathrm{Re}\,(P^*Q) = 0$; the second condition is the phase coherence.
- The bars show the palindromic form $P = \pm\frac{1}{\sqrt{2}}$, $Q = \pm\frac{i}{\sqrt{2}}$; $u \to -u$ and $u \to u\,R_z(\gamma)$ change nothing.

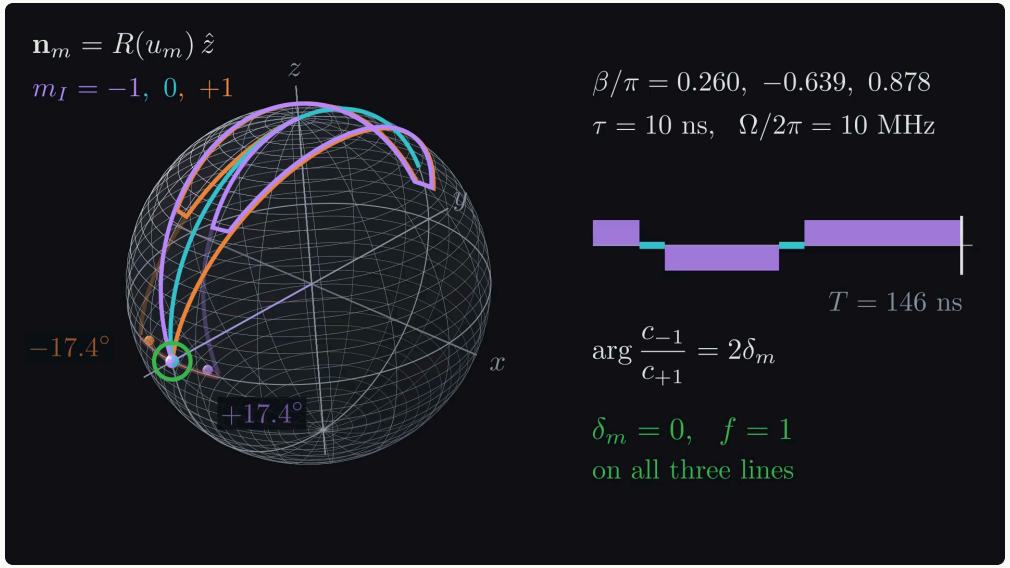
METHOD

DQ preparation: finite pulses

A hard $\pi/2$ pulse prepares the target at every Δ; a finite one leaves a DQ phase error.

$$U_{\text{finite}}\left(\tfrac{\pi}{2}\right) \simeq R_z(\Delta t_e)\, R_x\left(\tfrac{\pi}{2}\right) R_z(\Delta t_e) \implies \frac{1}{\sqrt{2}}\left(e^{-i\Delta t_e}|{+1}\rangle + e^{i\Delta t_e}|{-1}\rangle\right), \qquad t_e = \frac{1}{\Omega_{\text{DQ}}}$$





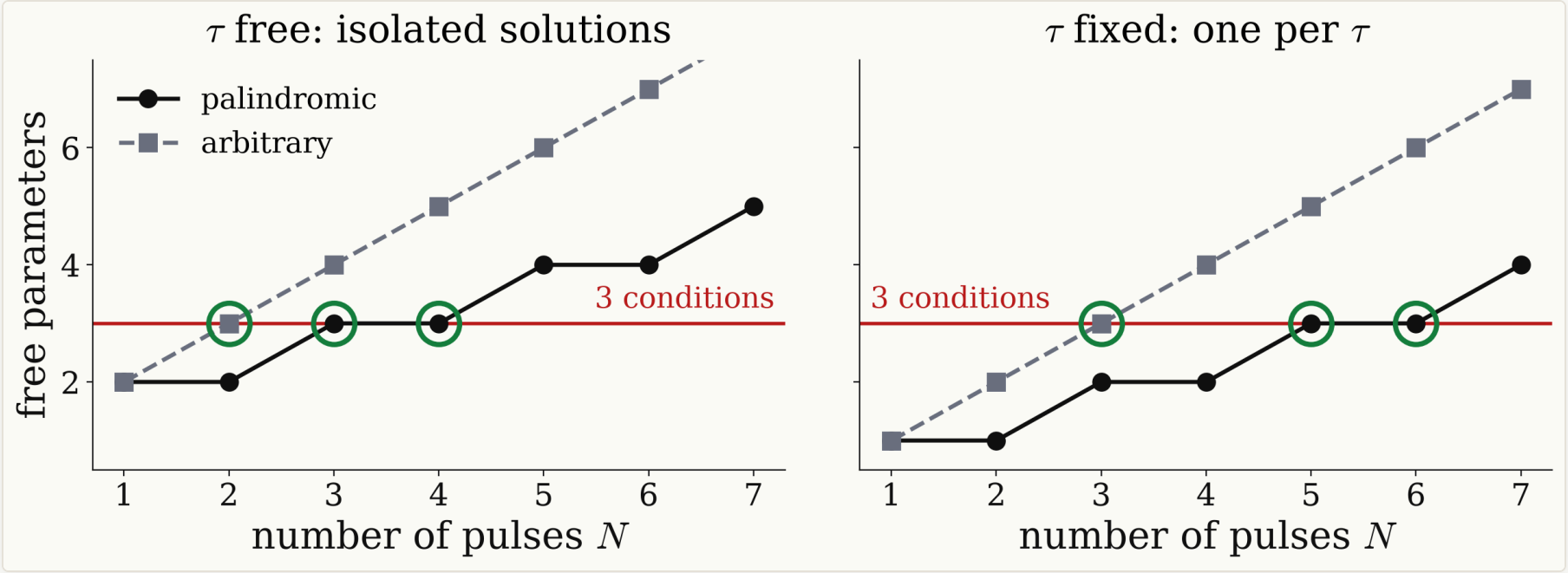
- With hard pulses the target is a constant: one Chebyshev coefficient, $K = 0$.
- One finite pulse gives $1 - f \simeq (\Delta/\Omega_{\text{DQ}})^2$: 2.3×10^{-2} (20 MHz) and 9.0×10^{-2} (10 MHz) on $m_I = \pm 1$.
- No wait can undo it (it would need negative time), so the pulses must.

METHOD

DQ preparation: exact structure

Finite pulses about $\pm x$ have three exact properties, which leave three conditions to solve.

1. Mirror symmetry: $u(-\Delta) = X\,u(\Delta)\,X$, so only $\Delta = 0$ and $\Delta = A$ need fitting.
2. Line centre: $u(0) = R_x\left(\sum_k \beta_k\right)$, so $\sum_k \beta_k \equiv \pi/2 \pmod{\pi}$.
3. Palindromic: $u^T = u \Rightarrow Q = -i\,r$ with r real, so coherence becomes $\mathrm{Im}\,P(A) = 0$.

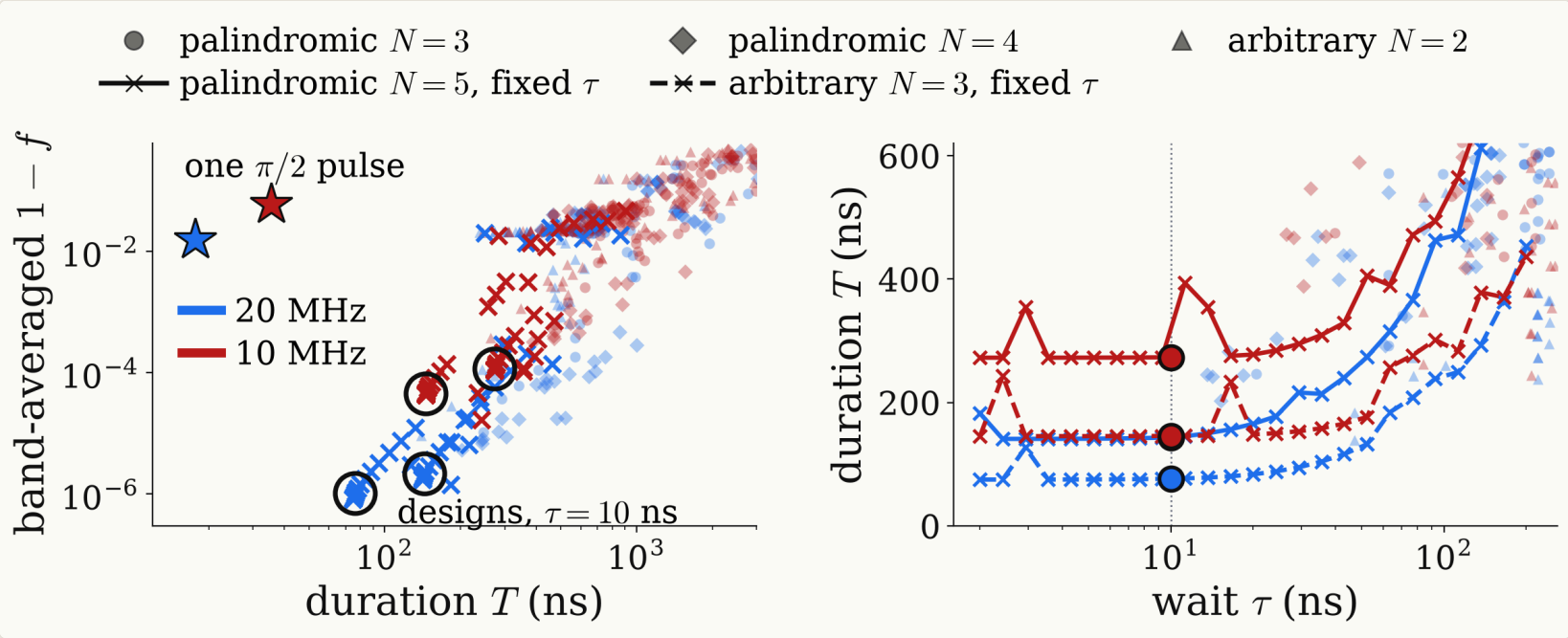


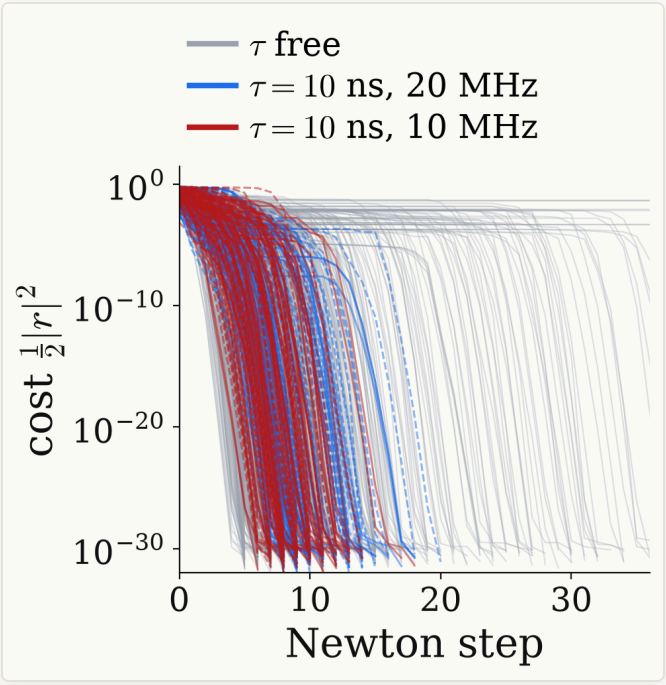
- Coherence costs no extra condition: the target is a state, so both classes need 3.

STRATEGY

DQ preparation: proper τ

One common target means no overlap rule; τ is chosen for a short sequence with a small band error.



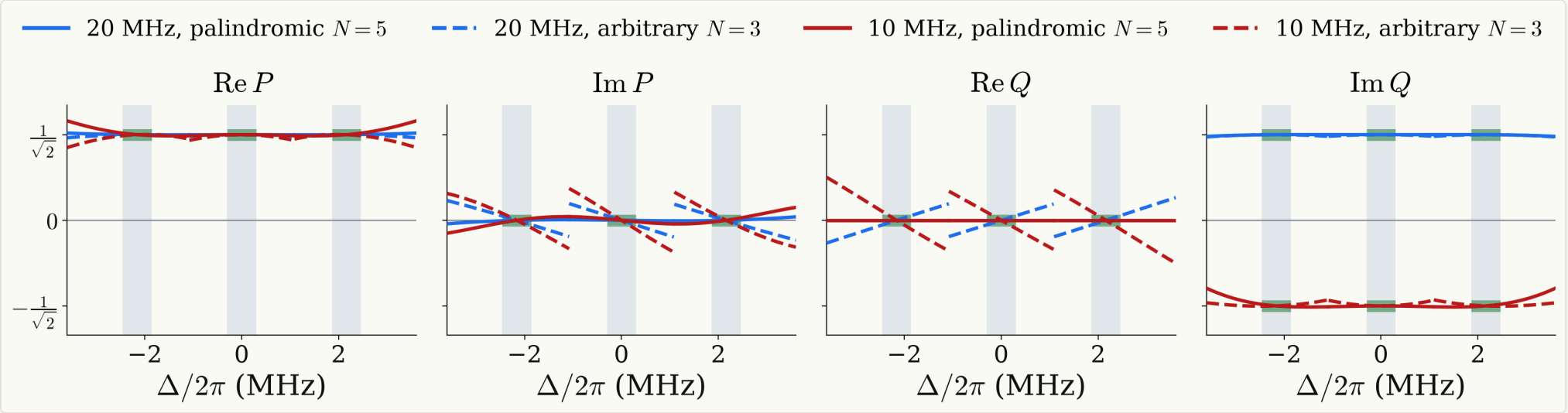


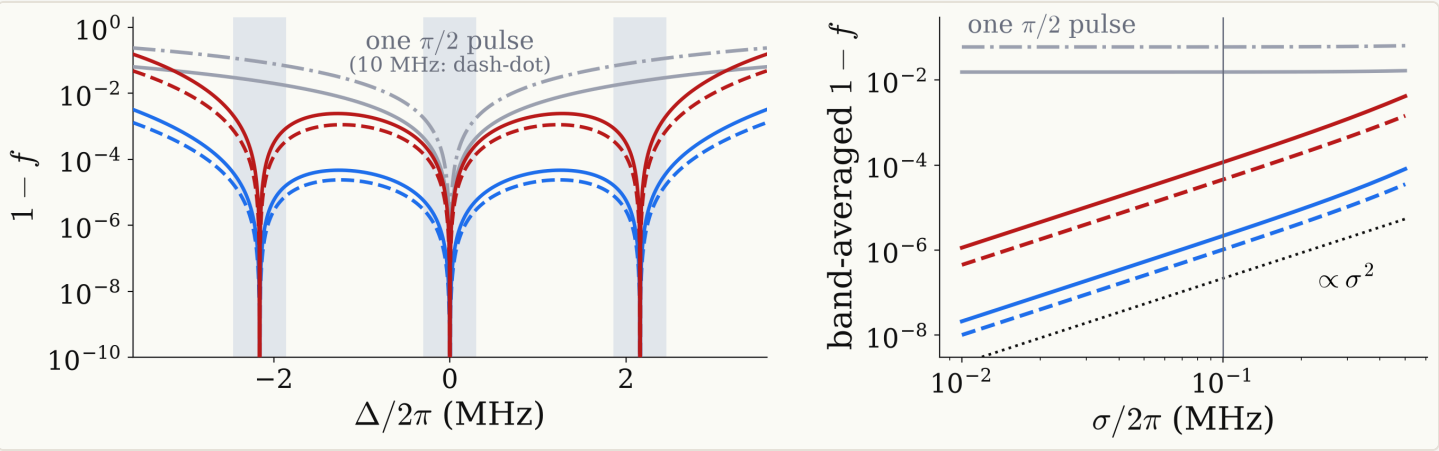
- For short τ the arbitrary family is shorter than every isolated solution, and both families have a smaller band error.
- Below $\tau \approx 10$ ns the duration hardly depends on τ (composite pulses as $\tau \to 0$), so we take $\tau = 10$ ns.
- At fixed τ, damped Newton converges from random starts in 6 to 20 steps.

RESULTS

DQ preparation: response

The four designs prepare the target exactly at the three line centres, with the same relative phase.



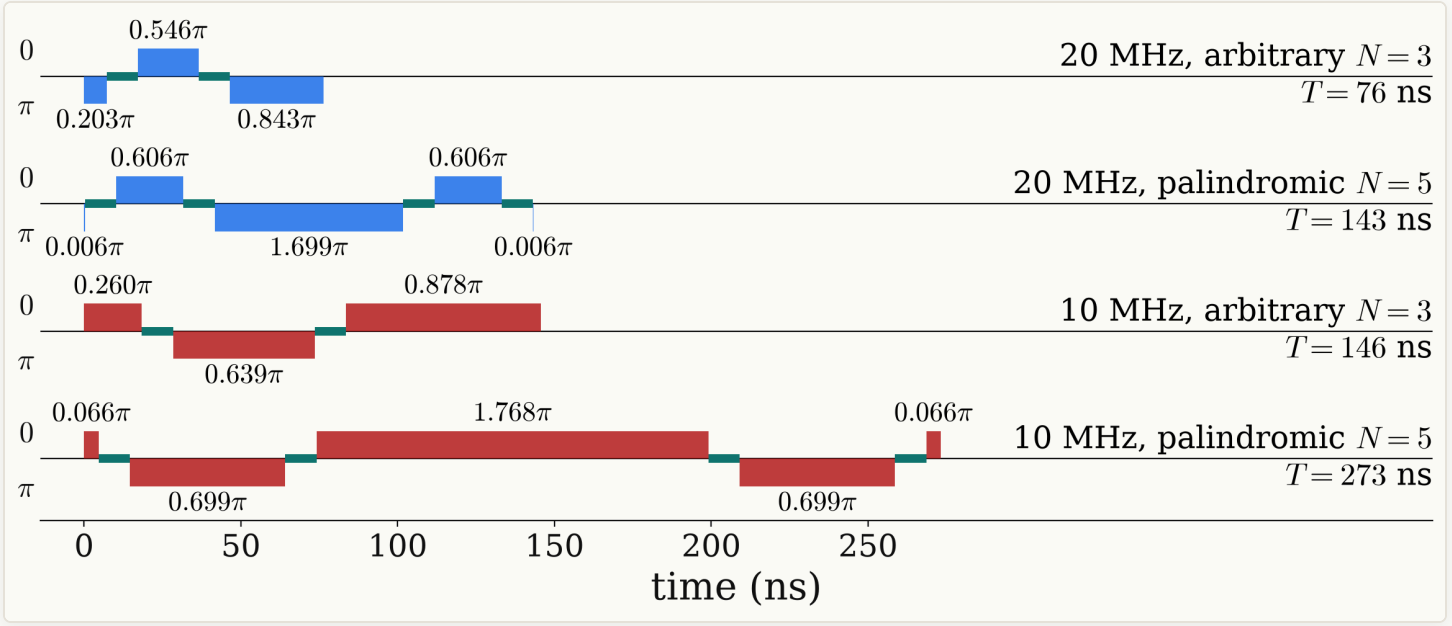


- The global phase of each manifold is fixed at its line centre.
- The relative phase $\arg(c_{-1}/c_{+1})$ is zero at all three centres.
- The band error grows as σ^2: exact at the centres, not flat.

RESULTS

DQ preparation: pulse sequence

At $\tau = 10$ ns the arbitrary sequence is the better one: three pulses, half as long.

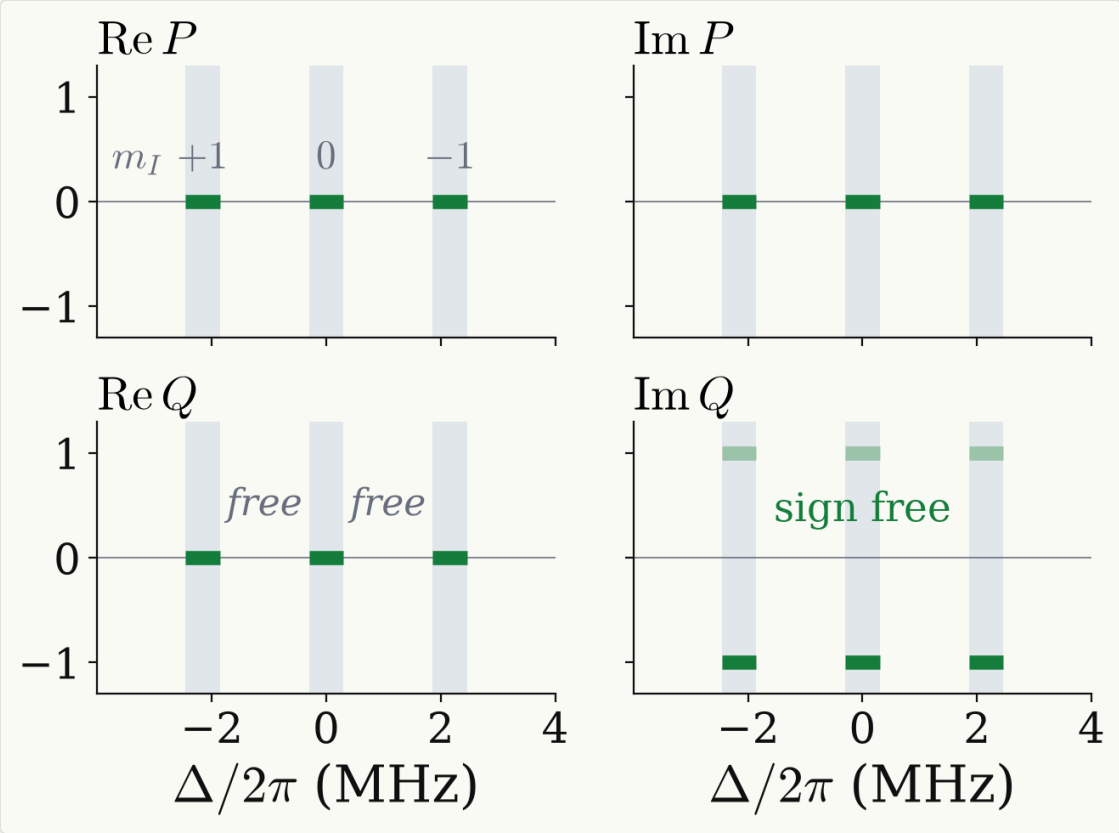


$\Omega/2\pi$	sequence	β_k/π (negative: phase π)	T	band $1-f$
20 MHz	arbitrary $N=3$	$-0.203,\ 0.546,\ -0.843$	76 ns	1.0×10^{-6}
20 MHz	palindromic $N=5$	$-0.006,\ 0.606,\ -1.699,\ 0.606,\ -0.006$	143 ns	2.2×10^{-6}
10 MHz	arbitrary $N=3$	$0.260,\ -0.639,\ 0.878$	146 ns	4.6×10^{-5}
10 MHz	palindromic $N=5$	$0.066,\ -0.699,\ 1.768,\ -0.699,\ 0.066$	273 ns	1.2×10^{-4}

STRATEGY

DQ π pulse: target function

The DQ π pulse swaps $|+1\rangle$ and $|-1\rangle$ with the same spin-1 unitary on all three lines.



$$P = 0: \qquad D^1(u) = \begin{pmatrix} 0 & 0 & Q^{*2} \\ 0 & -1 & 0 \\ Q^2 & 0 & 0 \end{pmatrix}$$

The same on every line iff $Q_m^2 = Q_0^2$: target $P = 0$, $Q = \mp i$ on the bands.

$$F = \frac{(4t^2 - 1)^2 + 3}{12}, \qquad t = |\mathrm{Im}\, Q|$$

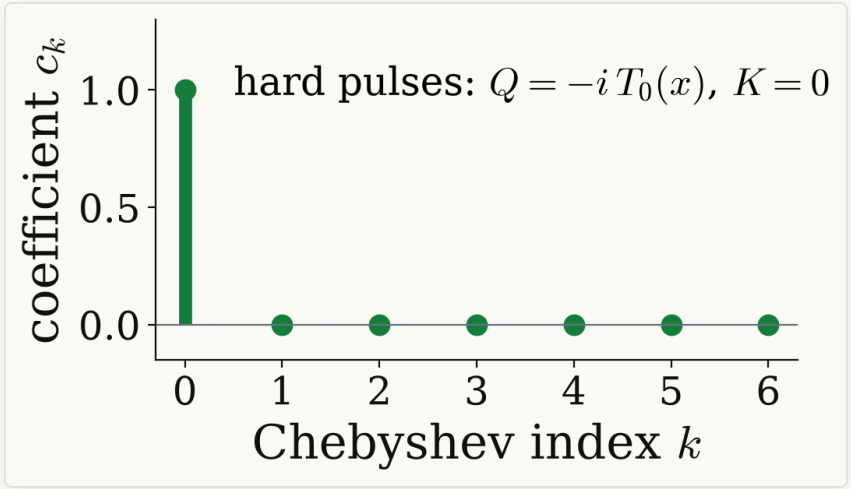
$$1 - F \simeq 2\left(|P|^2 + (\mathrm{Re}\, Q)^2\right)$$

- A common U has $U S_z U^\dagger = -S_z$, so $e^{-i\Delta S_z t}\, U\, e^{-i\Delta S_z t} = U$ for every Δ.
- A train of these pulses refocuses the hyperfine and static ^{13}C shifts.

METHOD

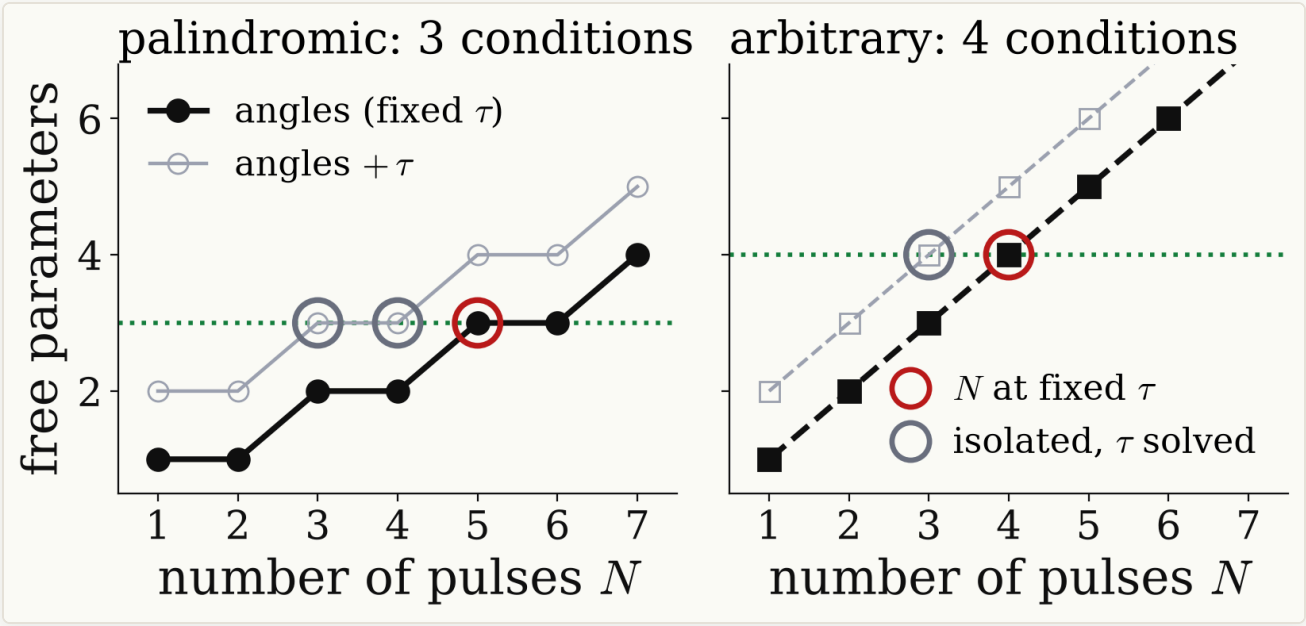
DQ π pulse: finite pulses

Hard pulses need one coefficient; for finite pulses, counting conditions gives the number of pulses.



$$U_{\text{finite}}(\pi) = R_x(\pi)\left[1 - i\,\frac{\Delta}{\Omega_{\text{DQ}}}\,Y\right] + O(\Delta^2)$$

- No free evolution undoes a y rotation: $1 - F = 4.6 \times 10^{-2}$ (20 MHz) and 0.17 (10 MHz) on $m_I = \pm 1$.



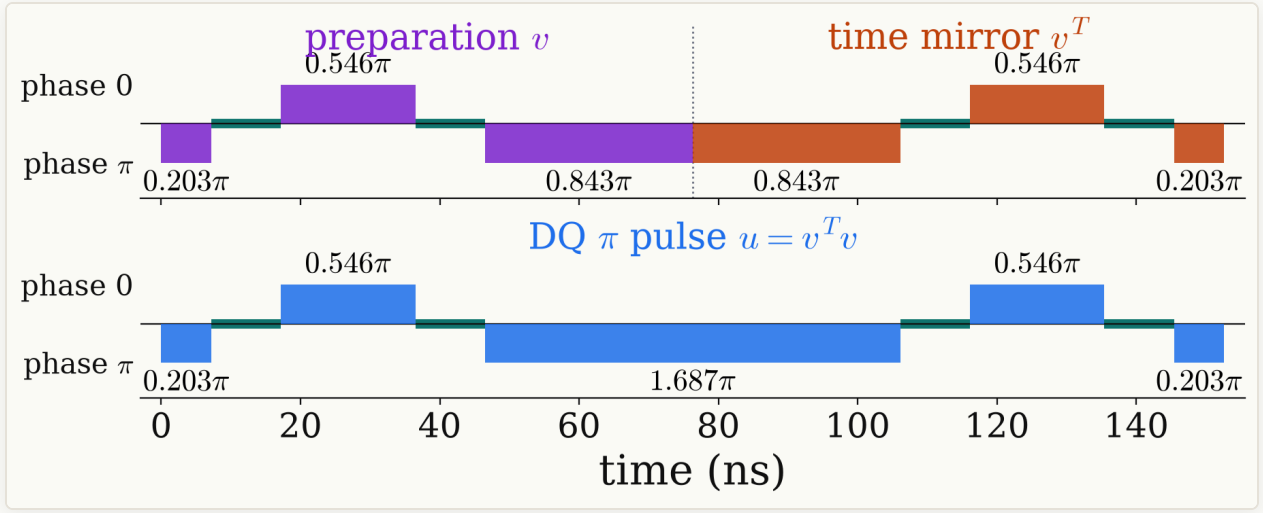
- The conditions at the centres are $\sum \beta \equiv \pi$ and $\operatorname{Re} P(A) = \operatorname{Im} P(A) = 0$, plus $\operatorname{Re} Q(A) = 0$ for arbitrary sequences.
- At fixed τ this needs $N = 5$ palindromic or $N = 4$ arbitrary pulses.

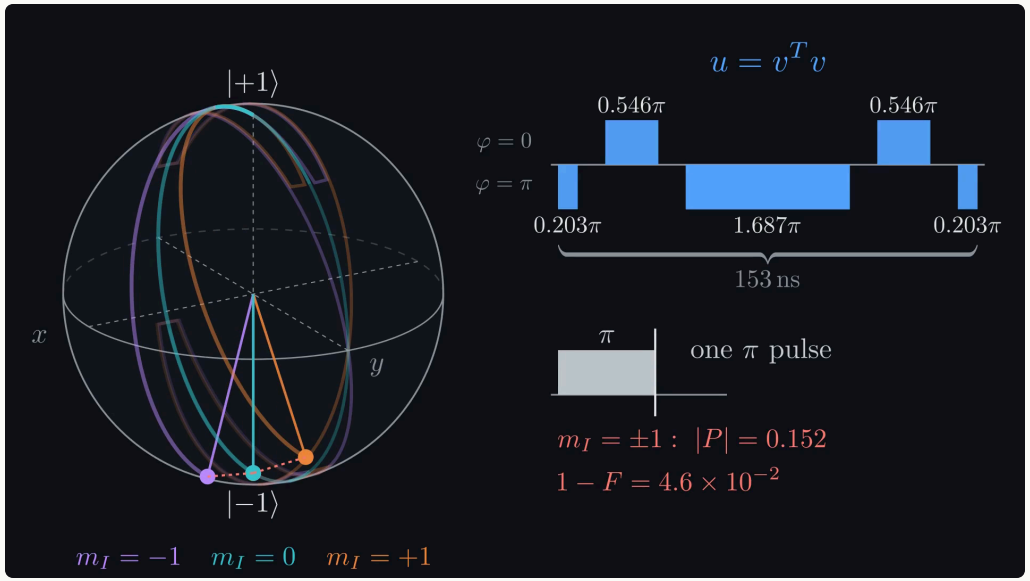
METHOD

DQ π pulse: time mirror

A preparation followed by its time mirror is a phase-coherent DQ π pulse.

$$v = \begin{pmatrix} p & -q^* \\ q & p^* \end{pmatrix} \xrightarrow{\text{reversed order}} v^T = \begin{pmatrix} p & q \\ -q^* & p^* \end{pmatrix}, \qquad u = v^T v: \quad P = p^2 + q^2 = 0 \ \text{ since } q = \pm ip$$



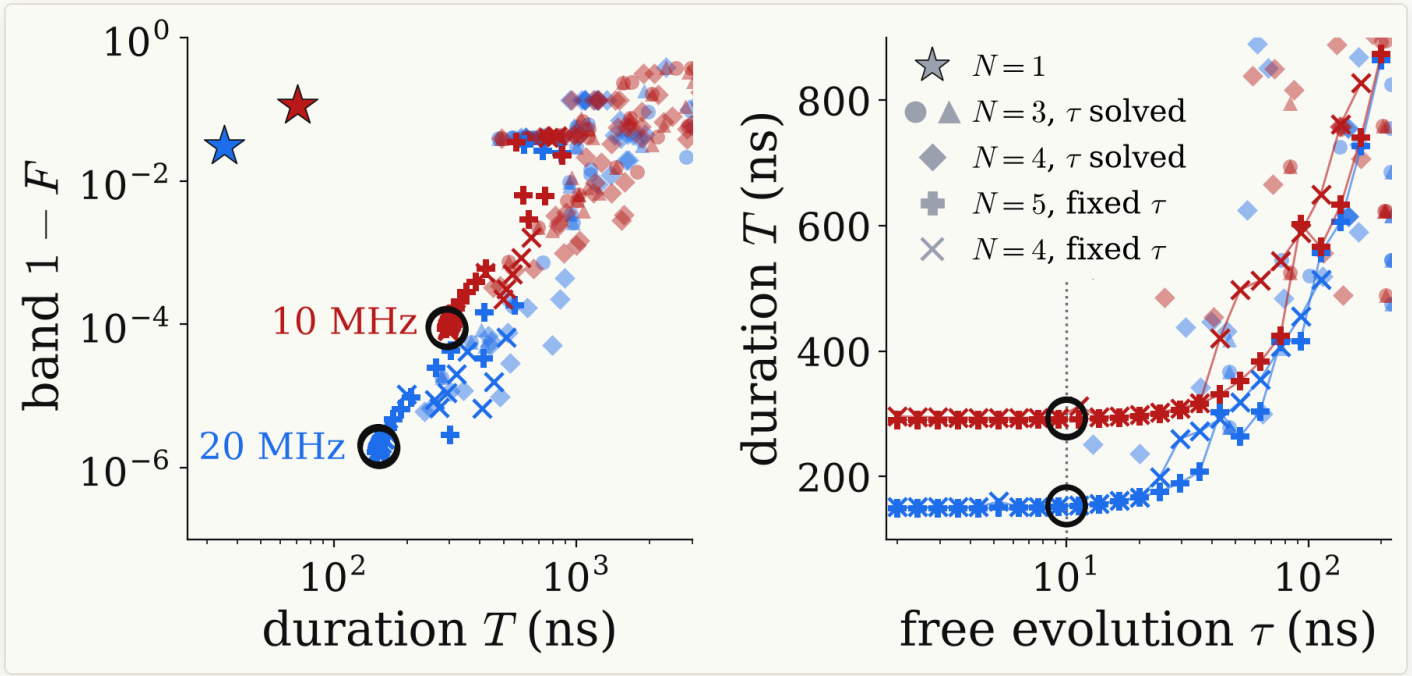


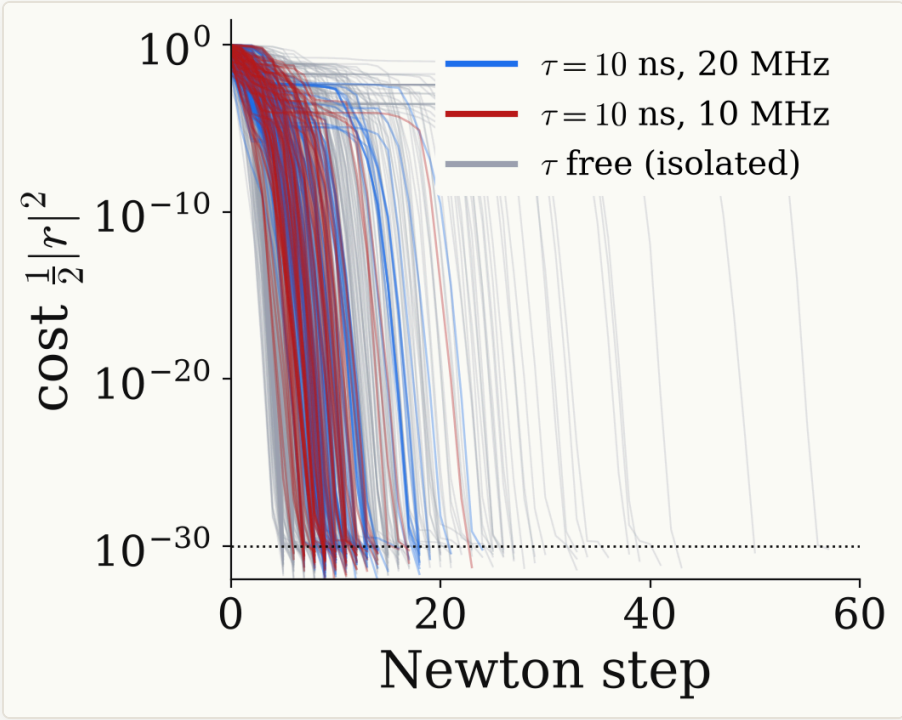
- At 20 MHz the mirrored preparation is the symmetric π design, 153 ns, twice the preparation.
- At 10 MHz it is the symmetric design with every pulse phase shifted by π, an equivalent gate.

STRATEGY

DQ π pulse: proper τ

At short τ the fixed-τ sequences are the shortest and most accurate; Newton iteration finds them in about ten steps.



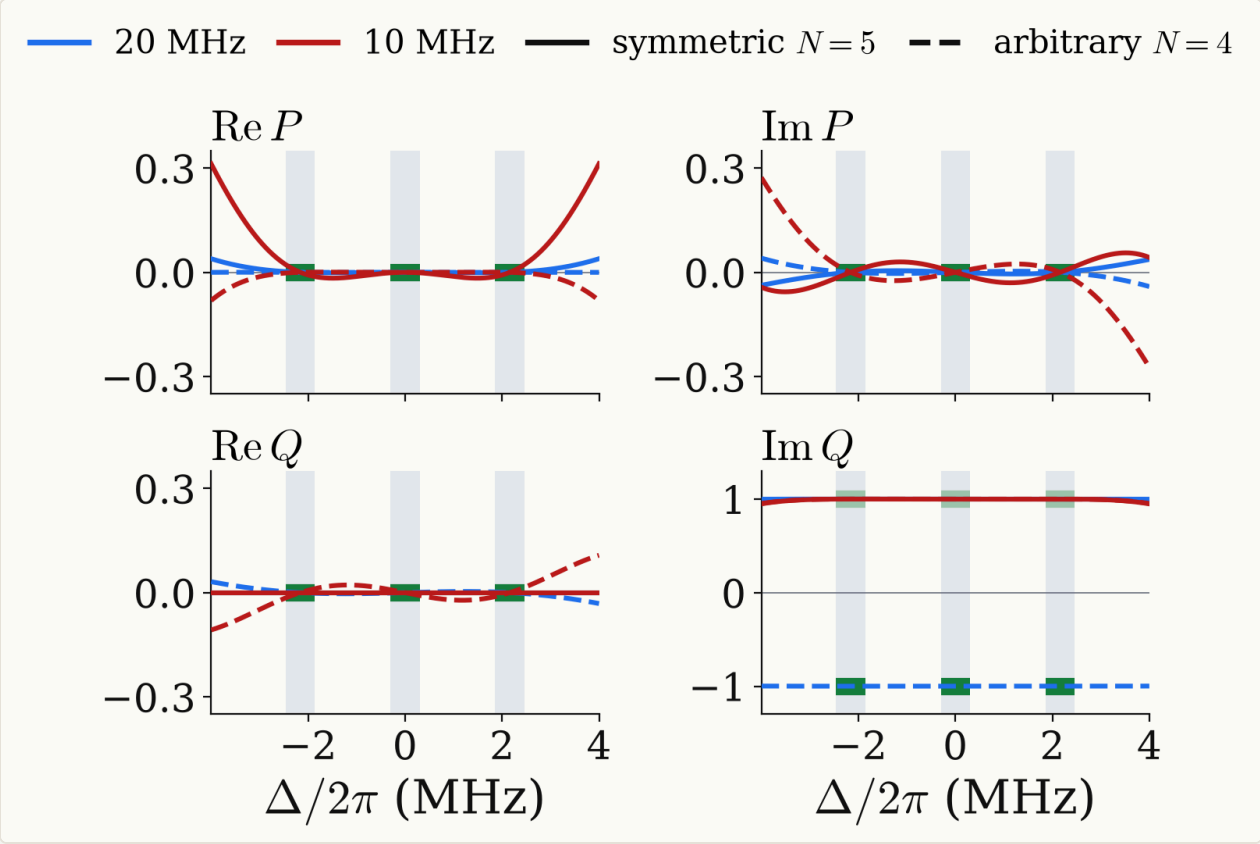


- Isolated solutions fix τ themselves and are longer: palindromic $N = 4$ at 20 MHz takes 236 ns ($\tau = 20.1$ ns).
- Below $\tau \approx 20$ ns the duration T hardly depends on τ; we keep $\tau = 10$ ns.

RESULTS

DQ π pulse: response

On the three lines every design meets the target; between and beyond them the response is free.

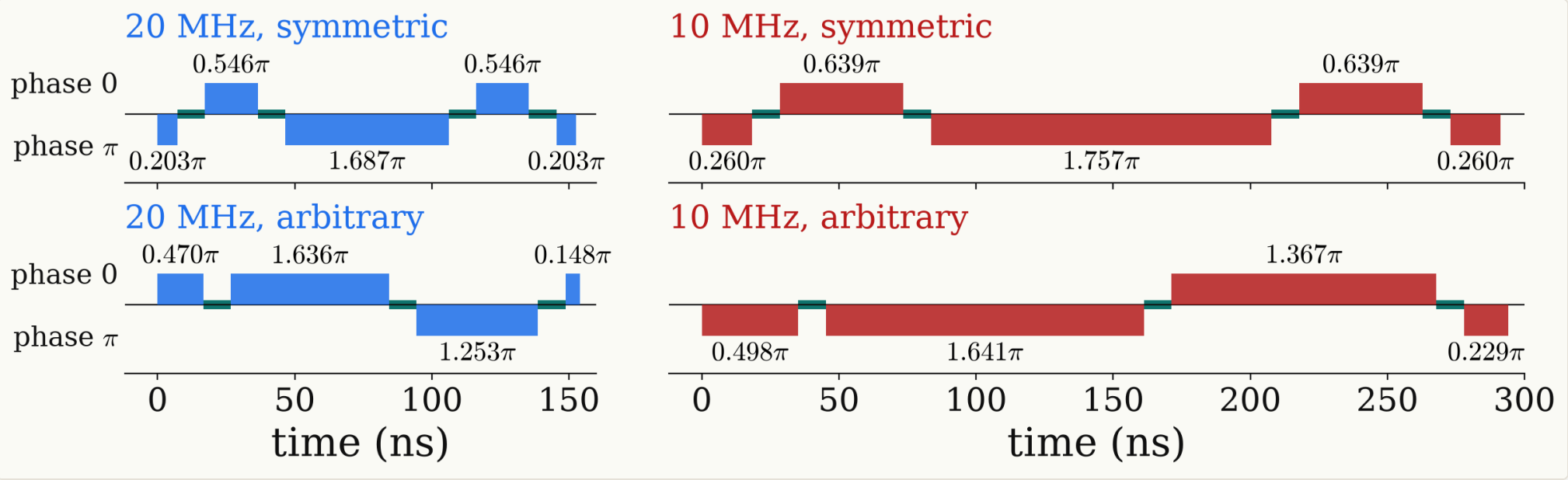


- The symmetric designs have $\mathrm{Re}\,Q \equiv 0$ at every Δ.
- The arbitrary designs have $\mathrm{Re}\,Q = 0$ only at the line centres.
- Away from the lines P grows, faster at 10 MHz.

RESULTS

DQ π pulse: pulse sequence

Four or five full-amplitude pulses with $\tau = 10$ ns invert all three lines with one phase.



$\Omega/2\pi$	design	β_k/π (negative: phase π)	T	band $1-F$
20 MHz	symmetric, $N=5$	$-0.203,\ 0.546,\ -1.687,\ 0.546,\ -0.203$	153 ns	2.1×10^{-6}
20 MHz	arbitrary, $N=4$	$0.470,\ 1.636,\ -1.253,\ 0.148$	154 ns	1.9×10^{-6}
10 MHz	symmetric, $N=5$	$-0.260,\ 0.639,\ -1.757,\ 0.639,\ -0.260$	291 ns	9.1×10^{-5}
10 MHz	arbitrary, $N=4$	$-0.498,\ -1.641,\ 1.367,\ -0.229$	294 ns	8.6×10^{-5}

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Thank you

Questions & discussion

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B

Backup

The Pauli form, designs exact at the three resonances, a wider stopband, real-pulse training, degree scaling

BACKUP

Pauli form

The Pauli form of a $\pm x$ sequence holds in every basis.

$$u = A(x)\,\mathbb{I} - iB(x)\,X - i\sin\theta\,\big[C(x)\,Z + D(x)\,Y\big], \qquad A^2 + B^2 + (1-x^2)(C^2+D^2) = 1$$

A, B real, degree $\le K$, parity K; C, D real, degree $\le K-1$, parity $K-1$ (K waits).

	Z basis	$\pm$ basis
stay	$P = A - i\sin\theta\, C$	$P_\pm = A - iB$
flip	$Q = -iB + \sin\theta\, D$	$\langle -\lvert u\rvert + \rangle = -i\sin\theta\, Q_\pm,\ \ Q_\pm = C - iD$
unitarity	$\lvert P\rvert^2 + \lvert Q\rvert^2 = 1$	$\lvert P_\pm\rvert^2 + (1-x^2)\lvert Q_\pm\rvert^2 = 1$

- A wait maps $(A, B, C, D) \to \big(xA - (1-x^2)C,\ xB - (1-x^2)D,\ A + xC,\ B + xD\big)$.
- $u(-\Delta) = X\,u(\Delta)\,X$, so $P(-\Delta) = P(\Delta)^*$ and $Q(-\Delta) = -Q(\Delta)^*$.
- A palindromic sequence has $u^T = u$, so $D = 0$ and $Q = -iB(x)$.

BACKUP

Example ($K = 2$)

Multiplying out $u = W\,R_x(\phi)\,W$ in both bases checks unitarity; $c = \cos\frac{\phi}{2}$, $s = \sin\frac{\phi}{2}$.

Z basis.

$$u = \begin{pmatrix} z^{-1} & 0 \\ 0 & z \end{pmatrix} \begin{pmatrix} c & -is \\ -is & c \end{pmatrix} \begin{pmatrix} z^{-1} & 0 \\ 0 & z \end{pmatrix} = \begin{pmatrix} c\,z^{-2} & -is \\ -is & c\,z^{2} \end{pmatrix}, \qquad |P|^2 + |Q|^2 = c^2 + s^2 = 1$$

$\pm$ basis. $R_x(\phi) = \mathrm{diag}(e^{-i\phi/2}, e^{i\phi/2})$, $W = \begin{pmatrix} x & -i\sin\theta \\ -i\sin\theta & x \end{pmatrix}$:

$$P_\pm = \cos^2\theta\, e^{-i\phi/2} + (-i\sin\theta)^2 e^{i\phi/2} = c\,T_2(x) - is, \qquad Q_\pm = c\,U_1(x)$$

$$|P_\pm|^2 + (1-x^2)|Q_\pm|^2 = c^2\left[T_2^2 + (1-x^2)U_1^2\right] + s^2 = c^2 + s^2 = 1$$

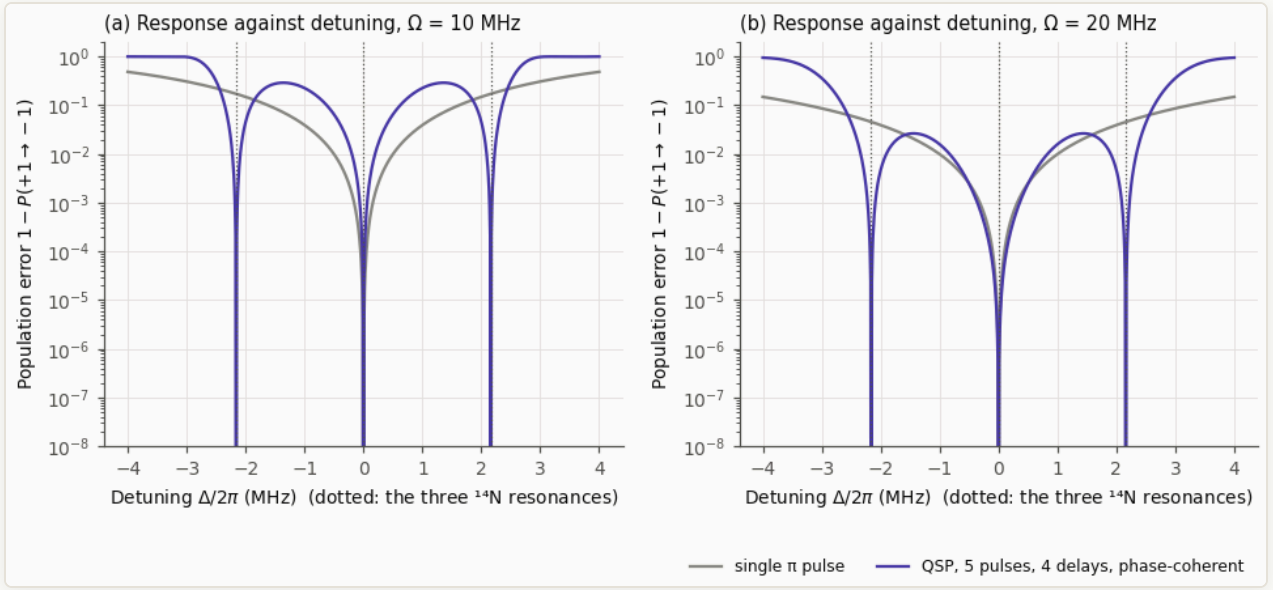
- The bracket is $T_n^2 + (1-x^2)U_{n-1}^2 = \cos^2 n\theta + \sin^2 n\theta = 1$.
- $q = |Q|^2 = s^2$ at every Δ.

BACKUP

Exact designs

Earlier designs are exact at the three resonances only.

$$G_1\,W\,G_2\,W\,G_3\,W\,G_2\,W\,G_1, \qquad \tau = \frac{2\pi}{3A}$$



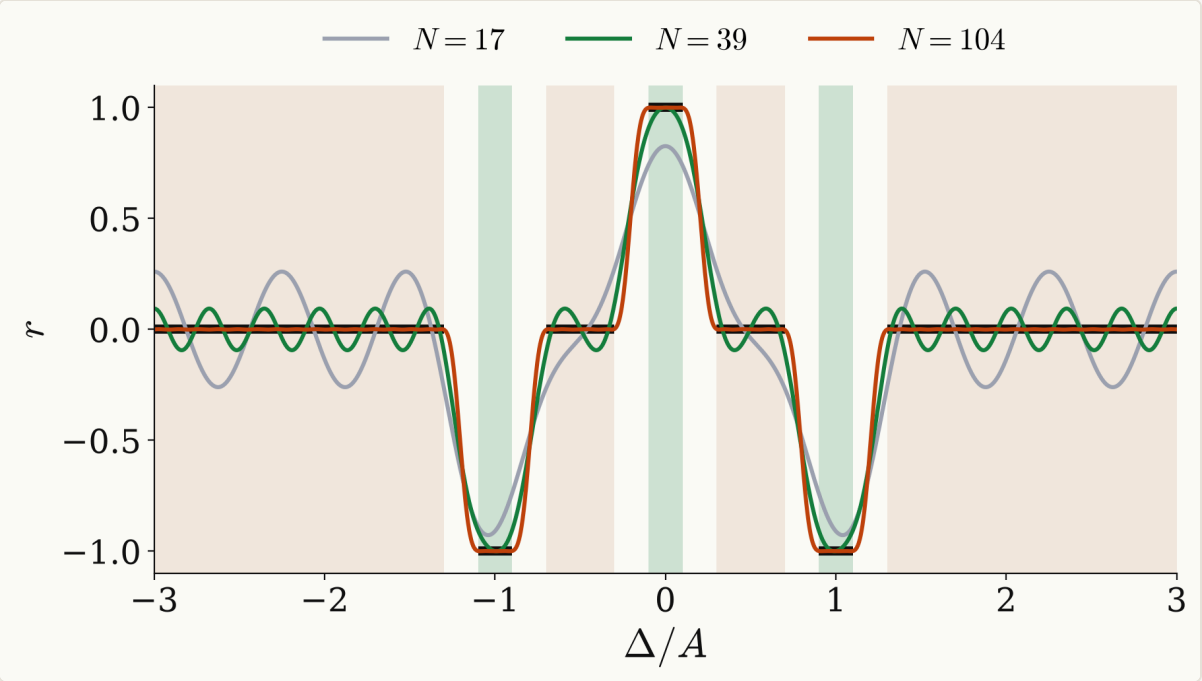
drive	fixed duration	full amplitude
Mamin et al.	777 ns	646 ns
20 MHz	971 ns	683 ns
10 MHz	1324 ns	747 ns

- Five palindromic pulses, phase-coherent.
- Nine-level model, one π pulse to design:
  $0.11 \to 3.1 \times 10^{-4}$ (10 MHz),
  $3.0 \times 10^{-2} \to 8.3 \times 10^{-4}$ (20 MHz).

BACKUP

Wide stopband

A stopband over $|\Delta| \le 3A$ needs 104 pulses for $E = 10^{-3}$.



$$\tau = \frac{2\pi}{6A}$$

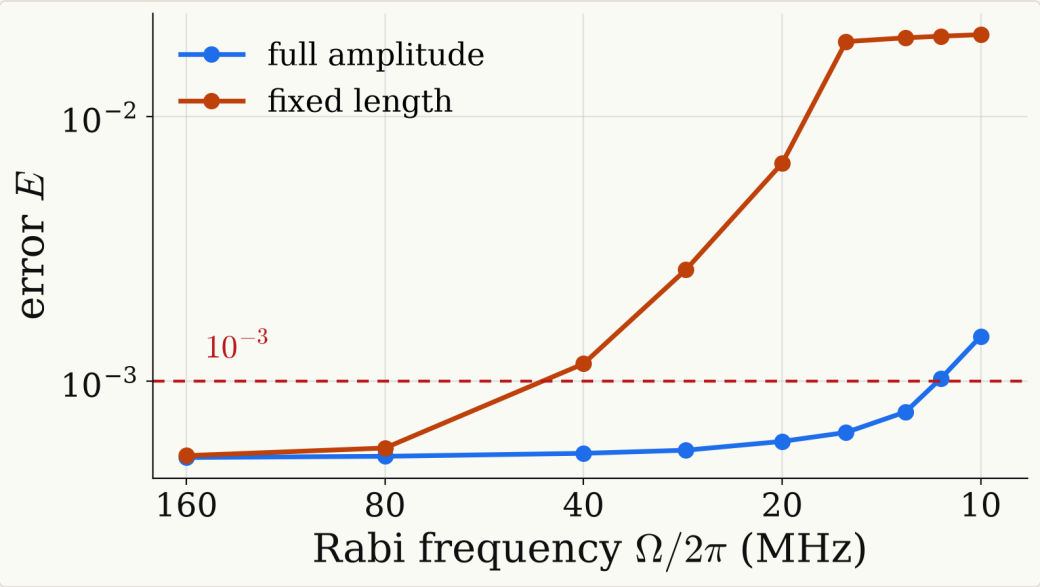
E	N	$K\tau$
0.1	36	2.7 μs
10^{-2}	68	5.2 μs
10^{-3}	104	7.9 μs

- At $\tau = 2\pi/3A$ the smallest degree is $K = 39$; $K\tau$ stays between 5.9 and $6.8\,\mu$s for every τ.
- Smoothed step series in $w = e^{i\Delta\tau}$: degree 48 instead of 21 at $2\pi/3A$, 109 instead of 41 at $2\pi/6A$.

BACKUP

Continuation in the Rabi frequency

The error grows smoothly as the drive weakens.

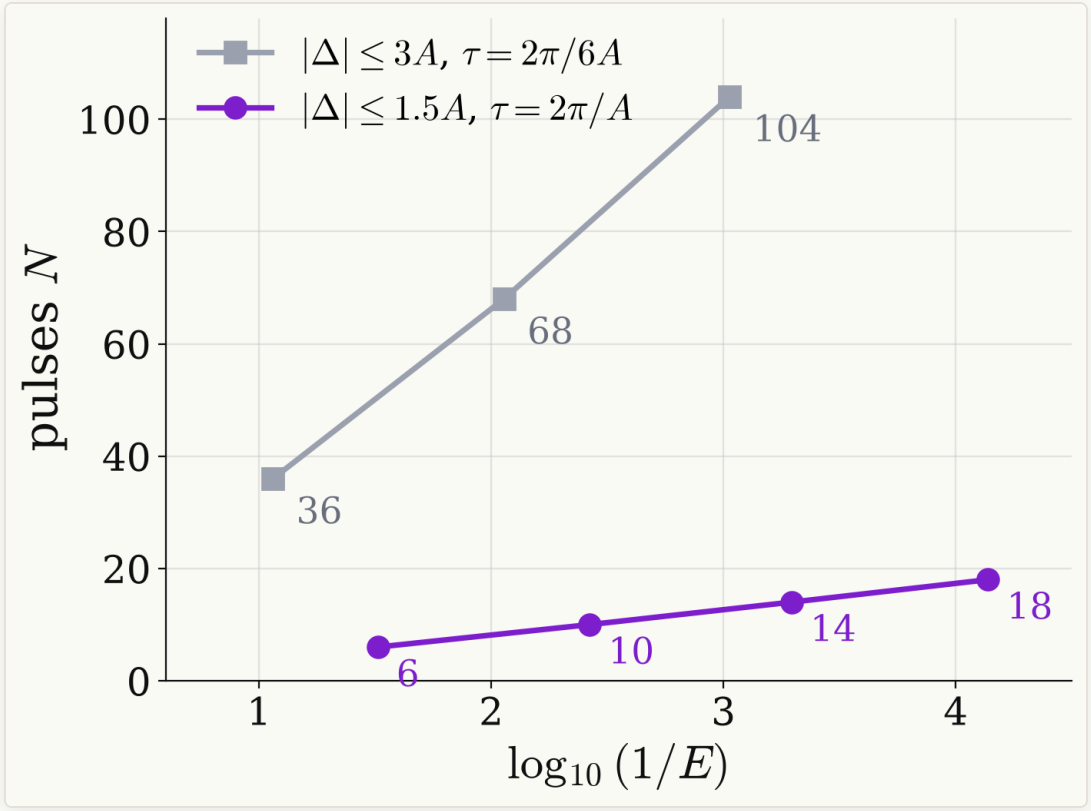


10 MHz, symmetric	$N = 14$ $(2\pi/A)$	$N = 27$ (π/A)	$N = 40$ $(2\pi/3A)$
full amplitude	1.5×10^{-3} (5.8 μs)	5.2×10^{-4} (5.9 μs)	5.2×10^{-4} (5.9 μs)
fixed length	1.4×10^{-2} (5.7 μs)	7.7×10^{-4} (5.8 μs)	2.6×10^{-4} (6.4 μs)

BACKUP

Degree scaling

A step across a gap needs degree $K \propto \ln(1/\varepsilon)/\Delta\theta$.



$$K \simeq c\,\frac{\ln(1/\varepsilon)}{\Delta\theta}$$

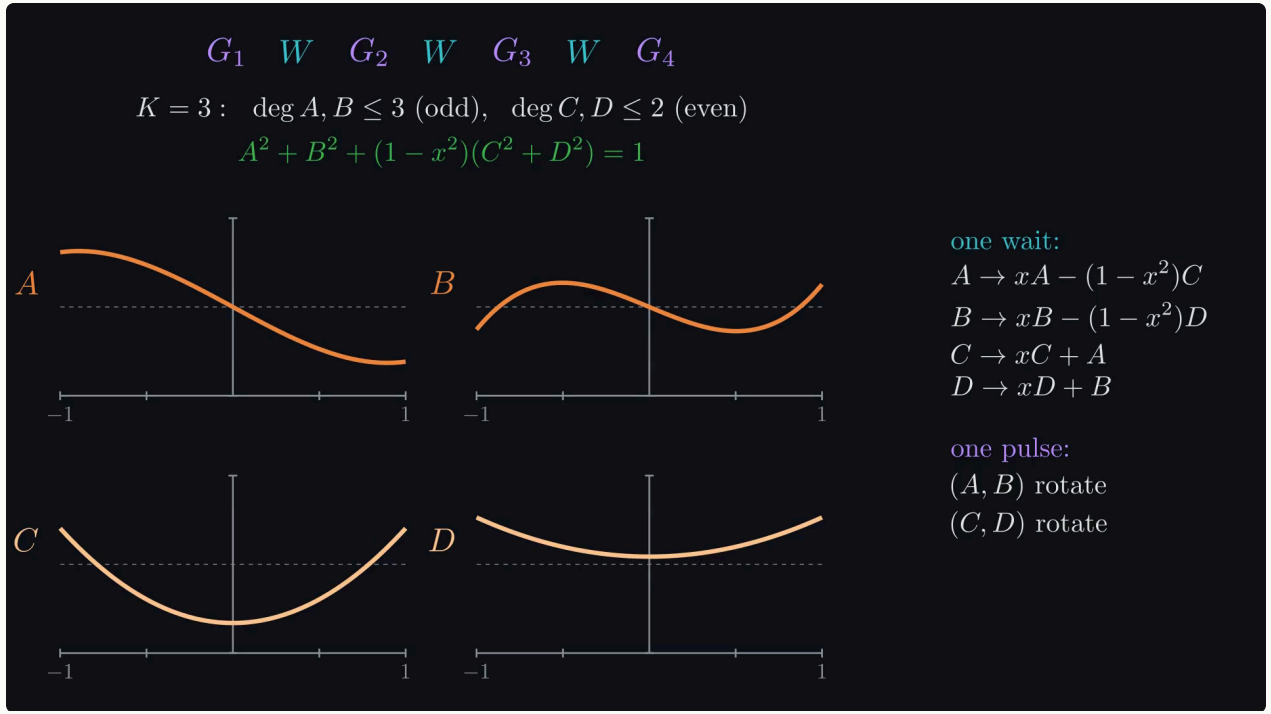
- The gap $\Delta\theta$ is the transition band, $0.2A$ in Δ: 0.2π at $\tau = 2\pi/A$, $0.2\pi/6$ at $\tau = 2\pi/6A$.
- Both targets need a number of pulses linear in $\ln(1/E)$.
- E_N falls by 7 to 8 per four pulses.

Low and Chuang (2017); Martyn et al. (2021)

METHOD Reachable polynomials

K waits and $K+1$ pulses about x make exactly these polynomials:

$$u = A\,\mathbb{I} - iB\,X - i\sin\theta\,(C\,Z + D\,Y), \qquad W = x\,\mathbb{I} - i\sin\theta\,Z, \qquad R_x(\phi) = \cos\tfrac{\phi}{2}\,\mathbb{I} - i\sin\tfrac{\phi}{2}\,X$$



real polynomials	degree	parity
$A,\ B$	$\le K$	of K
$C,\ D$	$\le K-1$	of $K-1$

$$A^2 + B^2 + (1-x^2)(C^2+D^2) = 1$$

$$P = A - i\sin\theta\,C, \quad Q = -iB + \sin\theta\,D$$

Conversely, every such A, B, C, D is made by $K+1$ pulses about x:

$$\underbrace{2(K+1)}_{\text{coefficients}} - \underbrace{(K+1)}_{\text{unitarity}} = \underbrace{K+1}_{\text{angles}}$$

Low, Yoder, Chuang, PRX 6, 041067 (2016); Martyn, Rossi, Tan, Chuang, PRX Quantum 2, 040203 (2021), Thm. 1